

PEPR SAMS – INTERFOOD PROJECT ANR-23-PESA-0004

Guidelines for encoding food items in the FoodEx2 system

Deliverable 1.1.1 Work Package 1

Version 3

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RELEASES:

Date	Released number	Main changes
2025/12	3	Modification of 3.5.2 How to code vegan or vegetarian foods?
2025/05	2	Addition of the Authors and Contributors page
2025/03	1	Creation of the document

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INTRODUCTION

Within the framework of the Priority Research Program and Equipment (PEPR) on Food Systems, Microbiome, and Health (SAMS), the project INTERFOOD focuses on the interoperability of food databases. Its objective is to enable an integrative approach to the various dimensions characterising foods: consumption, composition, contamination, environmental impact, price, etc.

This requires linking different food databases built according to heterogeneous nomenclatures. To date, matching food databases is done almost entirely manually, representing highly complex, time-consuming, and labour-intensive tasks. To create synergies and make progress on automating these matches, Anses, INSERM, and INRAE have joined forces through the INTERFOOD project. The first work package of this project aims to develop an IT tool to ease the link of two databases previously coded according to the EFSA¹ FoodEx2 nomenclature.

Indeed, to ensure the standardisation of the data it receives, EFSA requests the use of the Standard Sample Description (SSD2) data model, which includes the FoodEx2 system to describe food products (<https://www.EFSA.europa.eu/en/data/data-standardisation>). Many databases are already coded in FoodEx2 or will be in the future. Therefore, this nomenclature is a perfect candidate to serve as a matching pivot between different food databases.

In order to match two databases previously coded in FoodEx2, it is essential to code foods in a similar way. Based on the tools and guides provided by EFSA, this document aims to facilitate and standardise the coding of food databases in the FoodEx2 nomenclature to subsequently match them more easily using the tool developed by Anses as part of the INTERFOOD project. Its writing was carried out and coordinated by the two units "Food Observatory" and "Methodology and Studies" of the Risk Assessment Department of Anses.

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¹ European Food Safety Authority, <https://www.EFSA.europa.eu/fr>

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ABBREVIATION LIST:

PEPR: Priority Research Program and Equipment

SAMS: Food Systems, Microbiome, and Health

Anses: French Agency for Food, Environmental and Occupational Health & Safety

INSERM: National Institute of Health and Medical Research

INRAE: National Research Institute for Agriculture, Food, and the Environment

EFSA: European Food Safety Authority

FX2: FoodEx2

SSD2: Standard Sample Description

RPC: Raw Primary Commodities

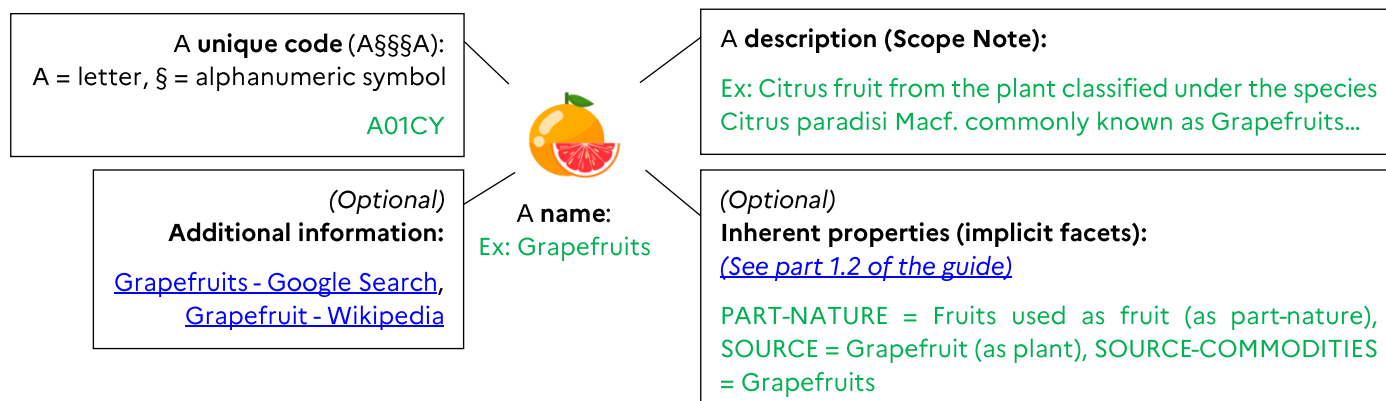
ONE PAGE TO UNDERSTAND FOODEx2

What is FoodEx2?

To ensure the standardisation of the data it receives, EFSA requires the use of the Standard Sample Description (SSD2) data model, which uses the FoodEx2 (FX2) system to describe food products. (<https://www.efsa.europa.eu/en/data/data-standardisation>).

The terms of the classification:

The FoodEx2 classification contains *terms* defined by:



Terms are organized into families within '**hierarchies**'. There are 9 different ones in the current version of FoodEx2.

[More information: part 1.3 of the guide](#)



Within a hierarchy, *terms* are categorised by their '**level of detail**' with five levels in the classification. The highest level includes *terms* representing broad groups of foods, while the lowest level contains very specific *terms*. Depending on its level of detail, a *term* may be more or less recommended for coding.

[More information: parts 1.1.2 and 2.2.6 of the guide](#)

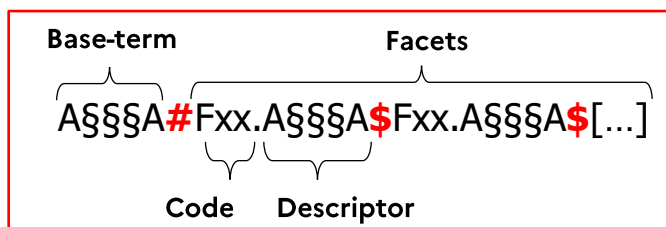


Terms are also categorised by their '**type**'. There are three types: raw primary commodities, derivatives of raw commodities, and composite foods.

[More information: part 1.1.3 of the guide](#)

FoodEx2 code:

In the FoodEx2 classification, any food item can be coded by combining *terms* from the classification within a single code, as shown below:



The **base term** is the main element of the FX2 code: it corresponds to a *term*, chosen from a **hierarchy** of the classification that best describes the food item to be coded.

[More information: part 2.2 of the guide](#)



Facets are used to further specify the *base term*, if necessary. There are 29 facets, each focusing on different characteristics (Ingredient, Production method, Alcohol percentage, Preservation method, Packaging, Processing methods, etc.). They consist of a **code** associated with a **descriptor**, drawn from a set of *terms*. The '**type**' of the *base term* may influence which facets are allowed.

[More information: parts 1.2 and 2.3 of the guide](#)

1 USEFUL DEFINITIONS AND TERMINOLOGY

1.1 ABOUT FOOD ITEMS

1.1.1 List terms and *base terms*

The FoodEx2 classification consists of a set of terms connected to each other in families, much like a family tree. Therefore, not all terms in the classification are intended for coding foods. Some are used only to classify and organise food items, such as the term "Fruit"; these are list terms.

A term used to describe a food product during data coding is called a **Base term**, the starting point of the FoodEx2 code.

1.1.2 Classification of the list terms according to their level of detail

List terms are categorised according to their « level of detail »:

- **Hierarchy terms:** These terms represent aggregated groups. They are used to regroup *base terms* during data analysis.

These terms should not be used for data coding.

"Fruits and fruit-based products" [A01BS] is a Hierarchy term.

- **Generic terms:** In general, these terms indicate a type of food, such as sauces or hot drinks, without being very specific about their nature. Whenever possible, **these terms should not be used for data coding** except in rare cases when precise information about the food is not available or more specific terms do not apply.

"Fruits used as fruits" [A04RK] is a Generic term.

- **Non-specific terms:** These terms are intermediate aggregated groups (for example: citrus fruits). **Their use should be limited** to cases where no Core term or Extended term correspond, and where the Non-specific terms provide more detail than the Generic term.

"Citrus fruits" [A01BT] is a Non-specific term.

- **Core terms:** These terms refer to foods or food families with sufficient precision for data coding. The goal is to code at this level or below the level of items classified as Core terms.

These terms can be used for data coding

"Grapefruit and similar" [A01CX] is a Core term.

- **Extended terms:** These terms provide more detail than a Core term.

These terms can be used for data coding.

"Grapefruit" [A01CY] is an Extended term.

1.1.3 Classification into 3 types of term

The list terms are also categorised based on their “type”. The FoodEx2 system distinguishes three types of food groups according to the food nature:

- **Raw Primary Commodities (RPC):** Pieces physically separated from a live source after harvesting plants or slaughtering animals, which may also be less invasive, like picking fruit or collecting milk or eggs. No changing nature process has been applied to the food. (Figure 1)



Figure 1: Raw carrot, RPC example

- **Derivatives of Raw Commodities:** Food products obtained from raw commodities where processes altering the ‘nature’ of food have been applied. A list of these processes, which modify the nature of the food, is provided later in this report, see 2.2.3. (Figure 2)



Figure 2: Carrot juice, derivative example

(The juicing process altered the carrot nature)

- **Composite Food:** Food products involving recipes (association of different ingredients). (Figure 3)



Figure 3: Carrot cake, composite example

Note: There is a fourth category, '**Broad**' foods, which includes general terms that encompass terms from different types. These are rarely recommended for coding, but if a term from this category must be used, the rules to apply are those for composite foods.

1.2 ABOUT FACETS AND DESCRIPTORS

The **facets** refer to the various characteristics of a food item (cooking method, preservation method, fat content, processing, etc.).

Descriptors are associated with each facet and offer various options to detail the different characteristics (for example, the facet F28-Process includes descriptors for the cooking method aspect: "roasted," "boiled," "grilled," etc.).

Descriptors within the facets are hierarchically structured, akin to a tree arrangement rather than a simple list of items.

Facets descriptors can have two functions in the system: **Implicit facets** or **Added facets** (though the correct terminology should be Implicit facet descriptors and Added facet descriptors)

- **Implicit facets** are facet descriptors that always and obviously apply to a specific food group. They are pre-filled in FoodEx2 and the user coding a food item does not need to take care of them.
- **Added facets** are facet descriptors inserted in the code while coding a food item. They describe characteristics of the specific food item making it slightly different from others, within the same food group.

Example:

Code for « Organic orange juice in a plastic bottle »:

A03AM#F18.A07NM\$F19.A07PR\$F21.A07SE

The term chosen as *base term* is « juice, orange » [A03AM]

This term has four implicit facets:

- A F02-Part-Nature facet descriptor indicating that the food is a ready-to-drink fruit/vegetable juice,
- A F27-Source-commodities facet descriptor indicating that the food comes from an orange,
- A F28-Process facet descriptor indicating that the food has been pressed,
- A F33-Legislative-classes facet indicating that the food belongs to the legal category FA-14.1.2 Fruit juice.

These facets do not appear in the code as they are implicit.

Additionally, three added facets descriptors have been added to the *base term*: the *Packaging-format* and *Packaging-material* facets, F18.A07NM and F19.A07PR respectively, specifying that the container is a plastic bottle and the *Production-method* facet descriptor, F21.A07SE, specifying organic agricultural production.

There are 29 facets in the current version of FoodEx2 (see Table 1 below).

The facets are described below, and their usage will be detailed later in the guide, see 2.3.

Table 1: Description of each facet

- Some of these facets describe the composition of the food

Code	Label	Description
F01*	Source	This facet describes the plant, animal, other organism or other source from which a raw primary commodity (RPC) has been obtained. In the case of plant and animals, the RPCs are parts separated from (e.g. meat, leaves, fruit...) or directly produced by (e.g. milk, eggs...) a source. <i>Ex: Garlic (plant), Goat (as animal), Yeast cultures (as organism)...</i>
F02	Part-Nature	This facet describes the nature of the food item or the part of plant or animal it represents. <i>Ex: Live cereal plants (as part-nature), Groats (as part-nature), Starch (as part-nature), Live mammals (as part-nature), Fat tissue (as part-nature)...</i>
F04*	Ingredients	This facet draws up a list of ingredients and/or flavour notes. It serves the purpose of providing information on ingredients of a composite food being important from some point of view, like allergic reactions, hazards, but also aspect, taste. <i>Ex: Milk, Chocolate</i>
F08	Sweetening-agent	This facet allows providing information on the added ingredient(s) used to give sweetness to a food item. <i>Ex: Sugars, Honeys, Syrups, Polyols, Artificial sweeteners</i>
F09	Fortification-agent	This facet allows providing information on the added ingredient(s) used to fortify a food item. <i>Ex: Vitamins, Calcium, Omega-3, Fibres, Caffeine, Probiotic</i>
F27*	Source-commodities	This facet describes the RPC from which an ingredient or derivative has been obtained. <i>Ex: Barley grains, Fishes, Vegetables, Milk, Tea</i>

*F01, F04 and F27 descriptors are similar but do not apply to the same type of term (RPC, derivative or composite). See 2.3.5.

- Others provide information on the origin of the food

Code	Label	Description
F21	Production method	The facet production method describes the method used to produce the food. It is mainly applicable for animals and for foods from plant and of animal origin. <i>Ex: Farmed / cultivated / aquaculture, Wild or gathered or hunted, Battery caged, Organic production</i>
F29	Purpose-of-raising	This facet allows recording the purpose of farming, keeping or breeding. <i>Ex: Meat production purpose, Milk production purpose, Breeding purpose</i>
F30	Reproductive-level	This facet allows recording classes of animals from the point of view of reproduction. <i>Ex: Elite breeding animal, Parent breeding animal, Animals not for breeding</i>
F31	Animal-age-class	This facet allows recording the classes of the animal used in legislation or in the practice, based on age or development stage. <i>Ex: Young mammal less than 1 year old, Hatching egg, Juvenile - immature, Adult</i>

F32	<i>Gender</i>	This facet records the sex status of an animal or group of animals. <i>Ex: Female, Pregnant female, Castrated male</i>
F34	<i>Host-sampled</i>	This facet provides information on the host-sampled indicating the host involved in the sampling process, i.e. for samples taken from vectors it indicates the species of the host where the vector was attached at the time of sampling (New 2024 facet). <i>Ex: Mammals, Birds, Environment</i>

- Additional facets offer further details about the food:
 - Some provide nutritional precision:

Code	Label	Description
F07	<i>Fat-content</i>	This is a facet with numerical descriptors, to provide the fat content (as percentage w/w) of a food item. <i>Ex: < 0.1 % fat, 6.7 % fat, 53 % fat, 100 % fat</i>
F10	<i>Qualitative Info</i>	This facet provides some principal claims related to important nutrients-ingredients, like fat, sugar etc. <i>Ex: Light, Sugar free, Gluten free, Sour, Red, Old, Integral, Sparkling, Not diluted</i>
F11	<i>Alcohol-content</i>	This is a facet containing information from the food label. This is a facet with numerical descriptors, to allow providing the alcohol (ethanol) content (as percentage v/v) of a food item. <i>Ex: < 0.1 % alcohol v/v, 6.7 % alcohol v/v, 100 % alcohol v/v</i>
F12	<i>Dough-Mass</i>	This facet is proposed to provide information on the original dough-mass, for bakery products. <i>Ex: Pizza dough, Short pastry dough (pate brisee), Choux pastry dough</i>

- Some focus on the physical characteristics of the food:

Code	Label	Description
F03	<i>Physical-state</i>	This facet describes the form (physical aspect) of the food as reported by the consumer (as estimated during an interview or as registered in a diary - Consumption Data) or as expressed in the analysis results in the laboratory (Occurrence Data). <i>Ex: En morceau, En tranche, En gelée, Pâte, Liquide, Poudre</i>
F06	<i>Surrounding-medium</i>	This facet is intended for food packed in any container, together with any additional (usually fluid) medium. It describes the nature of the medium surrounding the food. <i>Ex: In broth, In vegetable oil, In wine, In cream or milk</i>

- Others describe the processing and production methods:

Code	Label	Description
F17	<i>Extent-of-cooking</i>	This facet describes the intensity of heat treatment applied to a food item in the categories meat, fish/seafood, vegetables, eggs, bread and similar. <i>Ex: Half-cooked, Meat inside "raw", Vegetables inside raw, Egg yolk semi-liquid</i>
F18	<i>Packaging format</i>	This facet is used for packaged food and allows recording the container or wrapping form. <i>Ex: Box, Jar, Cup/pot, Capsule</i>
F19	<i>Packaging-material</i>	This facet is used for packaged food and allows recording the material constituting the packaging containing the food. In case of combined material, it describes all the material, not only the part in contact with food. <i>Ex: Edible material, Wood, Rubber, Aluminium, Plastic</i>
F22	<i>Preparation-product-place</i>	This facet allows recording the place where the food was prepared for consumption. <i>Ex: Food industry prepared, Restaurant, Prepared by artisan, Home prepared (family, social networks, proxies)</i>
F28	<i>Process</i>	This facet allows recording different characteristics of the food: preservation treatments a food item underwent, technological steps or treatments applied while producing a food item, the way a food item has been heat-treated before consumption and the way a food item has been prepared for final consumption (particularly needed for consumption surveys). This facet includes preparation (like battering or breading) as well as heat treatment steps. <i>Ex: Cooking and similar thermal preparation Processes, Physical division (Pulping / mashing, Slicing, Grating), Preservation treatments (thermal, Canning / jarring, Preserving with salt, with sugar) and other</i>

- While other provide information on safety and regulatory compliance:

Code	Label	Description
F25	<i>Risky-Ingredients</i>	This facet (of specific interest in the microbiological domain) allows recording the presence of microbiologically high-risk Ingredients. <i>Ex: Made from pasteurised milk, Containing raw eggs</i>
F33	<i>Legislative-classes</i>	This facet allows recording the food classes as reported in legislations. <i>Ex: FA-13.1 Foods for infants and young children</i>

- Another category includes general information about food consumption:

Code	Label	Description
F20	<i>Part-consumed-analysed</i>	When reporting food analysed or consumed, this facet allows specifying in which form the food item was analysed or consumed. <i>Ex: With skin, W/o shell, Only peel, With bone</i>
F23	<i>Target-consumer</i>	This facet allows recording different consumer classes intended as target for the food item. When reporting analyses for feed, a list or target animals is also included. <i>Ex: Infant or toddler food, Diabetics, Animal feed</i>

F24	<i>Intended-use</i>	This facet allows recording the intended use of a food item, in particular with respect to further treatment expected (or not expected) before consumption. <i>Ex: Ready-to-eat, Raw but intended to be eaten cooked</i>
-----	---------------------	---

- Lastly, there is a facet describing the accuracy of the FoodEx2 code:

Code	Label	Description
F26	<i>Generic-term</i>	This facet allows recording whether the food list code was chosen by default because of lack of information on the food item or because the proper entry in the food list was missing. <i>Ex: Unspecified, Other</i>

1.3 ABOUT HIERARCHIES

In the FoodEx2 system, food items are classified according to different hierarchies. These hierarchies represent different perspectives of the basic food list, see Table 2. Each term within a hierarchy is defined by a code, which remains consistent across all hierarchies, and its position within the hierarchy. Depending on the specific hierarchy, certain food items may or may not be available.

Two Hierarchies have to be used when reporting data:

- **The Reporting Hierarchy, for data reported to EFSA (such as contamination data);**
- **And the Exposure Hierarchy, for data used in exposition measures (such as consumption data).**

Table 2: List and description of hierarchies

Nom	Description
<i>MTX</i>	Collection of the full terminology (including facets) included in FoodEx2, useful to browse, especially the facet, but never to be used for coding
<i>Reporting</i>	Preferred hierarchy for reporting occurrence data to EFSA. This hierarchy eases the choice of the food code and helps the user to choose an enough detailed code (not only “fruits” or “cake” but “Apple” or “Sponge cake”).
<i>Zoonoses</i>	This hierarchy allows the coding of a broad variety of samples for the biological monitoring domain. The remaining food categories are organised as in the Reporting hierarchy.
<i>Feed</i>	This hierarchy includes all the food groups related to animal feeding. All terms contain the specification ('feed'), indicating that they are feed intended for animal consumption. To be used for animal feed data only.
<i>Exposure</i>	Preferred hierarchy for reporting consumption data and for assessing exposure. It is linked to food consumption therefore it only includes the food section.
<i>VetDrugRes</i>	This hierarchy encompasses the groups used for recording data on residues of veterinary medicinal products. In this domain, the data always refer to a specific animal, but may vary between food samples and non-food animal-related samples, such as biological samples or even feed samples, environmental samples or water used in farming.
<i>Botanicals</i>	This hierarchy has been created to support EFSA database of botanicals containing naturally substances of possible concern for human health when used in food and food supplements.
<i>FeedAddExpo</i>	This hierarchy is not usable. The FeedAddExpo hierarchy was developed to be used for the Feed Additives Consumer Exposure (FACE) calculator. This tool allows the estimation of dietary exposure to residues of feed additives.
<i>PRIMo</i>	This hierarchy is intended for the PRIMo (Pesticide Residue Intake Model), the standard tool employed at EU level for the dietary risk assessment for pesticide residues.

FoodEx2 considers facets as “attribute hierarchies”. The hierarchies presented above are “base hierarchies”.

1.4 UNIQUE CODES FOR LIST TERMS AND FACET DESCRIPTORS

To every element in the FoodEx2 terminology, whether it's a food item or a facet descriptor, is assigned a unique alphanumeric code consisting of 5 characters. The code itself holds no specific meaning and follows a consistent pattern: it begins and ends with a letter, with letters or numbers in between.

This code is unique because no other element in the terminology shares the same code. It remains unchanged across the different hierarchies and potential adaptations, including different versions of FoodEx2.

For example:

Food: A0B6Z

The facet codes (either for implicit or added facets) are reported with the following syntax:

- The code includes a header defining the facet (in the form 'Fxx.')
- And the unique term code for the descriptor.

Facets are never reported alone; they are always added to a *base term* in a complex code.

For example:

F28.A07JV

[F28] refers to the facet Process and [A07JV] is Smoking.

So, the code is translated as "PROCESS = Smoking".

1.5 ABOUT THE CATALOGUE FOODEx2 AND THE BROWSER

The classification of foods according to the FoodEx2 system and all the characteristics presented above (the level and categories of foods, definitions, hierarchies, etc.) are described in a document, the FoodEx2 catalogue, provided and updated by EFSA, in an Excel format (MTX file). See 4.2 for details on the Excel document.

EFSA developed and made available for all Member States an application allowing a visual representation of the food classification according to the FoodEx2 system: the **catalogue browser**. It employs a tree diagram to illustrate relationships between foodstuffs and facilitate the creation of codes in the required EFSA format.

This browser is available online on the EFSA website (<https://github.com/openEFSA/catalogue-browser/wiki>).

1.5.1 Main screen of the app

The main screen of the browser is divided into five panels (see Figure 4):

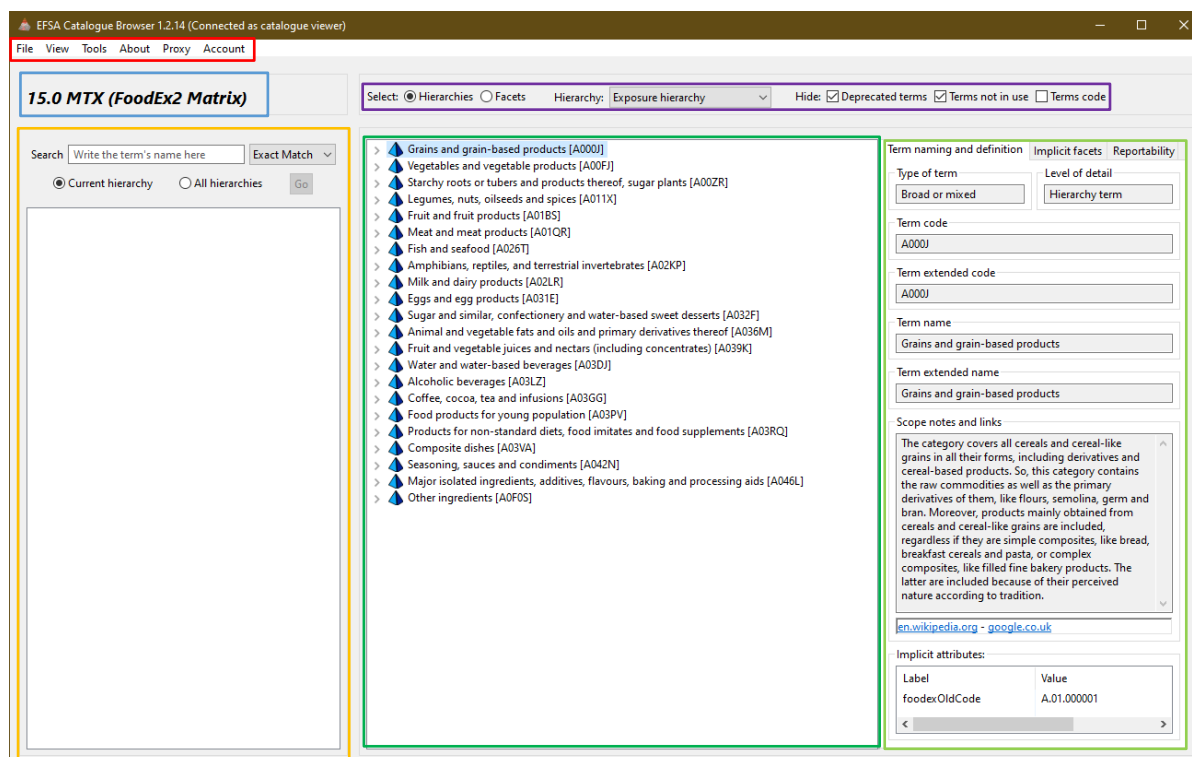


Figure 4: Overview of the catalogue browser

- **Main menu** (red rectangle): Contains all the drop-down options that allow the interaction between the software and the catalogues;
- **Catalogue name** (blue rectangle): Shows the catalogue's name (and the download button if updates are available);
- **Filter group** (violet rectangle): Can be used for filtering specific data inside the catalogue currently opened;
- **Search panel** (orange rectangle): Allows to search terms in the catalogue and then view the list of results in a special interactive section located just below;

- **Tree term panel** (green rectangle): Displays all the terms of the catalogue following a hierarchical tree view structure;
- **Information panel** (light green rectangle): Several tabs are shown on the right part of the screen in which it is possible to find information related to the selected term.

This interface provides a simplified method for strolling the FoodEx2 catalogue, making it easier for users to locate and access the desired information.

1.5.2 How to download a catalogue from the browser

Users with an EFSA account can connect with the 'Account/Log in with DCF'. Once connected, specific catalogues can be downloaded from the 'File' menu (in the Main Menu) (see Figure 5).

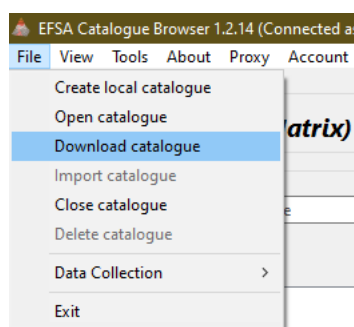


Figure 5: How to download a catalogue from the browser

Quite a few catalogues are available, including the MTX (FoodEx2 Matrix) versions, (see Figure 6).

Name	Version	Status	Last release	Scope
☐ MATRIX	6.1	PUBLISHED MINOR	2024-09-24	The ca
☐ MDACC (MDSTAT)	5.0	PUBLISHED MAJOR	2023-01-31	Catalo
☐ MTX (FoodEx2 Matrix)	15.2	PUBLISHED MINOR	2024-07-29	FoodE
☐ MTXCLS	6.0	PUBLISHED MAJOR	2024-01-26	Matrix
☐ MTXTYP	2.3	PUBLISHED MINOR	2024-02-05	Catalo
☐ Nuts Gaul	9.7	PUBLISHED MINOR	2024-09-10	The ca
☐ PARAM	15.30	PUBLISHED MINOR	2024-09-26	The PA
☐ PARAMCLS	5.4	PUBLISHED MINOR	2024-03-04	Param
☐ POP_SUBGROUP	7.0	PUBLISHED MAJOR	2023-01-30	Catalo
☐ POSNEG	4.0	PUBLISHED MAJOR	2024-01-25	Catalo
☐ PRGTYP	7.0	PUBLISHED MAJOR	2024-01-29	Catalo
☐ REF_TYPE	3.0	PUBLISHED MAJOR	2019-01-19	Scienti
☐ RISKORBENEFIT	1.0	PUBLISHED MAJOR	2023-08-11	This ca
☐ SAFETYNDIET	1.0	PUBLISHED MAJOR	2023-08-17	This ca
☐ SAMPI R	5.2	PUBLISHED MINOR	2024-03-25	Catalo

Figure 6: Example of different downloadable catalogues

Users without a DCF account (EFSA account) must first download the MTX FoodEx2 catalogue as an ".ecf" version from the [OpenEFSA GitHub \(https://github.com/openefsa/efsa-catalogues\)](https://github.com/openefsa/efsa-catalogues). This repository also contains all the other catalogues published by EFSA. After saving the file in a local folder, the catalogue can be imported into the catalogue browser via the option "Import Catalogue" of the "File" menu. A notification will be displayed when the process is successfully completed.

1.5.3 Overview of the details available in the browser

In the browser, it is possible to look for a term and its related information:

- In the filter group, the user can choose the preferred hierarchy. Then, the browser will display the corresponding tree structure (see Figure 7).

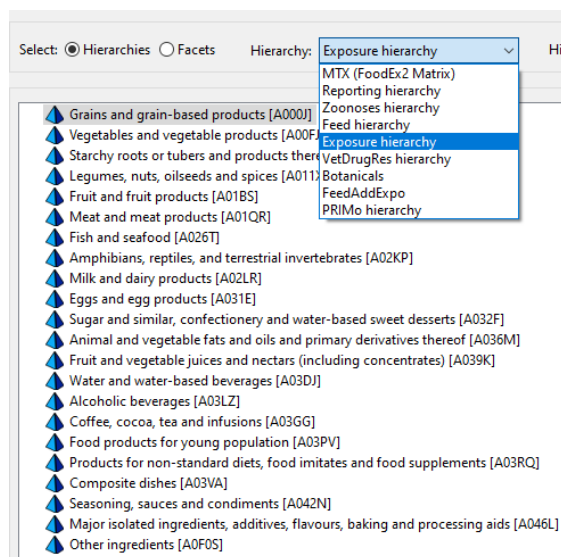


Figure 7: The 22 groups of the top panel of the **Exposure** Hierarchy in the catalogue browser

- Any item can be searched in the catalogue through their name or their unique code (see Figure 8). **Note** that the whole code of five characters must be provided in the search bar for the search to work. However, a search can be performed with an incomplete label (minimum 3 letters by default).

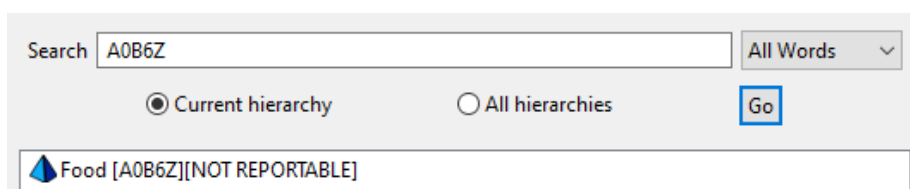


Figure 8: Search bar of the catalogue browser

- In the tree view, the level of detail is identified with different symbols and colours (see Figure 9).

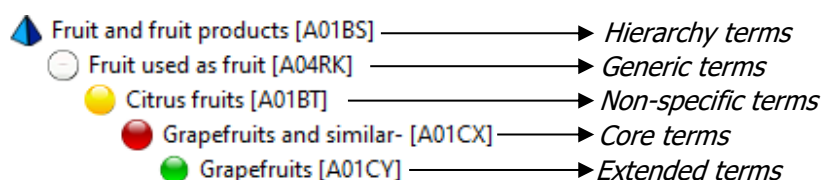


Figure 9: Symbols, colors and example for every level of detail in the catalogue browser

- In the information panel, the user can find for every term (see Figure 10):
 - The **level of detail**, available in the « Level of detail » field.
 - The **type of term**, in the « type of term » field.
 - The **list of implicit facets**, in the « Implicit facets » tab or as part of the term code in the « Term extended code » field.
 - The “**Scope notes and links**” field contains the definition of the term as well as links toward a Wikipedia page and a google images search link. Some terms don’t have links yet.
 - A list of implicit attributes is also given in the “**Implicit attributes**” field. It contains a number of additional information, including variations of the term name. When a search is done, the result shows terms with matching name but also matching descriptions or implicit attributes
 - The “**Reportability**” tab display all the hierarchies where the term is present and usable. It is particularly useful for confirming a term's presence across multiple relevant hierarchies.

Term naming and definition **Implicit facets** Reportability

Type of term
Raw Primary Commodities (RPC)

Level of detail
Extended term

Term code
A01CY

Term extended code
A01CY#F02.A0ELC\$F01.A05XY\$F27.A01CY

Term name
Grapefruits

Term extended name
Grapefruits, PART-NATURE = Fruits used as fruit (as part-nature), SOURCE = Grapefruit (as plant), SOURCE-COMMODITIES = Grapefruit

Scope notes and links
Citrus fruit from the plant classified under the species *Citrus paradisi* Macf., commonly known as Grapefruits. The part consumed/analysed is not specified. When relevant, information on the part consumed/analysed has to be reported with additional facet descriptors. In case of data collections related to legislations, the default part consumed/analysed is the one defined in the applicable legislation.

en.wikipedia.org - google.co.uk

Implicit attributes:

Label	Value
Scientific names	<i>Citrus paradisi</i> Macf.
GEMSCode	FC0203
matrixCode	P0110010-000
foodexOldCode	A.01.000546

Figure 10: Information panel for the term *Grapefruit* [A01CY] in the catalogue browser

2 GENERAL RULES OF CODIFICATION

The general method for coding in FoodEx2 is to:

1. [Choose the right base term](#)
2. [Choose appropriate facets descriptors](#)

Facets help to further define a base term, ensuring that relevant information is retained when the base term alone is insufficient.

If difficulties remain for some codifications despite this guide, the user can contact EFSA with the following email: data.collection@efsa.europa.eu.

2.1 REQUIRED STRUCTURE OF THE FOODEx2 CODE

The final code of FoodEx2 is a complex code including in a single string different pieces of coded information. It corresponds to the type 'compound data element' in the SSD2 (EFSA, 2013).

The FoodEx2 code consists of:

- the code of a *base term* (mandatory, otherwise the final code is not valid),
- Followed (optionally) by a hash character '#' and a sequence of facet codes separated by a dollar character '\$'

Examples:

1. A typical FoodEx2 code looks like the following:

A03BG#F09.A0EXH\$F10.A077L\$F21.A07SE

Nectar, orange #FORTIFICATION AGENT=Calcium\$QUALITATIVE INFO= Sugar free
\$PRODUCTION METHOD= Organic production

The \$ character is only used to separate facets.

2. Another example:

A03BG

Nectar, orange without any further detail.

To allow an automated management of the codes, the following requirements are added regarding the structure of the code:

- Spaces are not allowed in the FoodEx2 code;
- Only one FoodEx2 code is allowed per food item;
- Facets should always be put in alphanumerical order.

2.2 HOW TO START: FIND THE APPROPRIATE *BASE TERM*

After selecting the right hierarchy - see 1.3 - the first step is to find the right *base term*.

Selecting the appropriate *base term* is crucial in the FoodEx2 system as it categorises the term within a family of related food items. This placement is essential to consider for data analysis purposes later on. Sometimes, the choice of the family/category of the chosen term is as important as the code chosen.

Furthermore, choosing the correct *base term* already guarantees a good code, to a large extent. If additional details are needed, selecting the right *base term* first simplifies the process of deciding which facets to include.

2.2.1 What type of food groups? Choose the right starting point

The initial decision is whether to select as *base term* a raw commodity, a derivative or a composite.

The first important question in front of a food sample is therefore:

What type of food is this?

This question refers to the nature of the food itself and should never be confused with the question of the origin which would be: what is the food made from?

The main concepts with respect to the choice of the *base term* for a FoodEx2 code are summarised in Figure 11 and detailed in paragraphs thereafter. Furthermore, all these information are summarised in the ANNEX 3: BASE-TERM SUMMARY → RPC, DERIVATED OR COMPOSITE?, page 93.

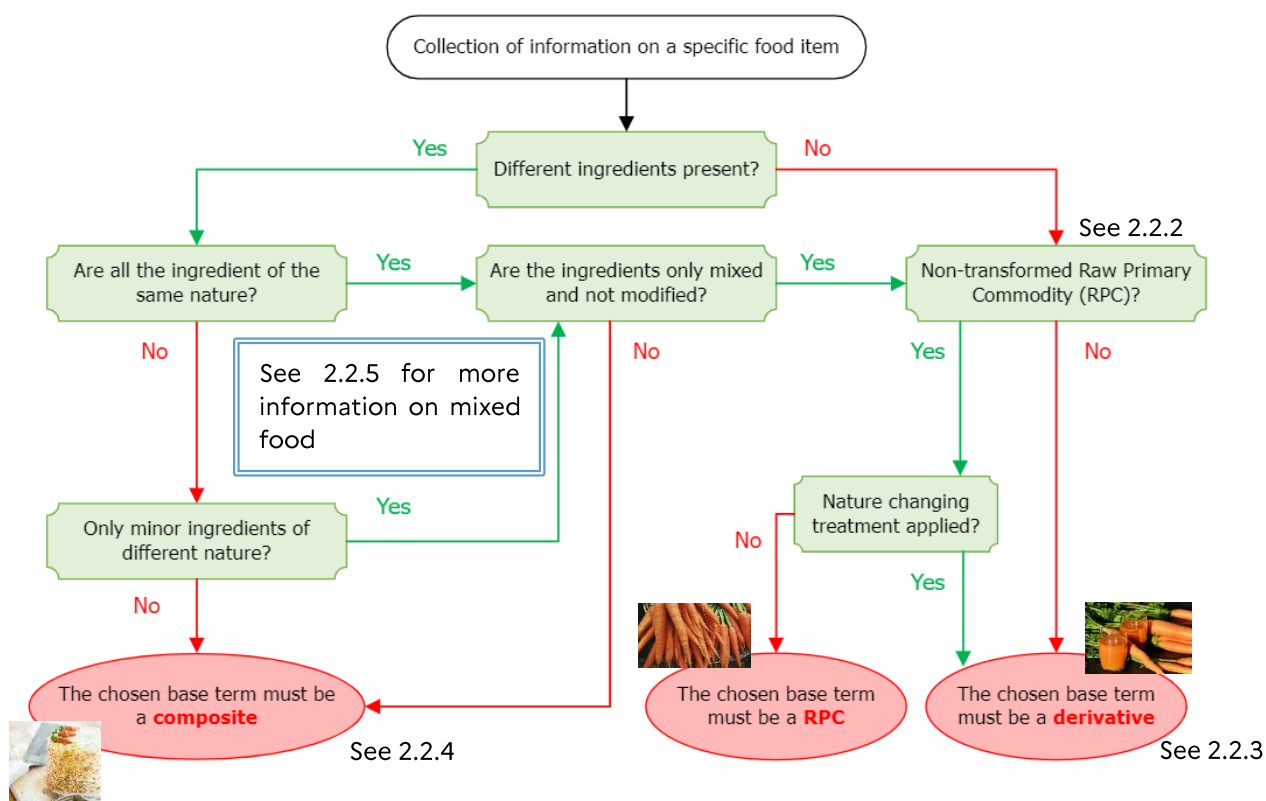


Figure 11: Decision tree for the choice of the *base term* for a specific food item.

The flowchart above is designed to assist coders in navigating the logic of the system and the recommended approaches. It outlines the process leading to the selection of the *base term* (RPC, derivative or composite). While the flowchart does not specify the level of detail (core term, extended term and so on), it directs users to the appropriate 'area' of the hierarchy. Here, coders make decisions between more generic or specific terms based on the available information.

2.2.2 The chosen *base term* must be a raw primary commodity

A raw commodity *base term* has to be chosen in all cases when the product is a primary unprocessed product from agriculture or animal husbandry.

For agricultural products normally available in both fresh and dried form, the **fresh form** is considered the raw commodity, while the **dried form** is a derivative; this applies to herbs, legumes and similar cases.

The products usually available in the dry form, because harvested as such, are considered raw commodity even in their dry form. However, if they are fresh, it must be specified through facet F10 – A166Y “QUALITATIVE-INFO = Fresh”.

Spices and herbal infusion materials are considered derivatives raw material (except tea since the amendment of 2016, now considered as raw commodity when fresh). See 3.6 for more information.

Example:

Typical examples of raw commodities are those listed in Table 3.

Table 3: Non-exhaustive list of food groups considered raw commodities in FoodEx2

Code	Food Group	Code	Food Group
A00KR	Leafy vegetables	A00VC	Wakame
A00KY	Head lettuces	A00VP	Spirulina
A00FV	Brussels sprouts	A0CVG	Royal jelly
A00YH	Sage	A031H	Duck eggs
A00FR	Cauliflowers	A01AR	Saffron
A010V	Sugar canes	A0CVS	Elk milk
A001K	Rye grain	A04HG	Cattle milk
A014H	Cashew nuts	A020Y	Bovine tail
A0BAL	Kidney bean (fresh seeds)	A01YC	Goose liver
A015M	Rape seeds	A01VA	Pig fat tissue
A01CS	Blood oranges	A0FCG	Giant river prawn
A01GN	Nectarines	A02EL	Cod liver
A01EL	Youngberries	A02DX	Tuna
A0CGD	Guavas	A0FAX	Dentex
A00QH	Carrots	A01SP	Chicken fresh meat
A01AC	Turmeric roots	A01QV	Bovine muscle
A00TG	Ceps	A04AV	Goat, minced meat

2.2.3 The chosen *base term* must be a derivative

An RPC derivative *base term* has to be chosen in all cases when the product is obtained from a raw commodity using one of the processes creating a new derivative nature, see Table 4.

In the FoodEx2 classification, there is a food type called "derivative", which includes raw materials that have undergone treatments altering their nature. However, not all treatments applied to a raw material result in the creation of a new term; in some cases, a descriptor from the F28-Process facet added to a raw primary commodity is enough. For example, grated carrots are coded with the *base term* "Carrots" [A00QH] and the facet descriptor F28.A07KZ "PROCESS = Grating" whereas carrot puree is coded with the derivative term "Vegetable puree or paste" [A0F3F] and the facet descriptor F27.A00QH "SOURCE-COMMODITIES = Carrots".

The decision on which derivatives of raw commodities to include, and which to simply describe with added facet descriptors, was complex. An approach, based on identifying the most relevant treatments for creating new natures and creating specific food groups for derivatives based on only these treatments, was preferred. The relevance of treatments was based on food science (expected change in composition and differences in terms of contamination) and observations of the market (different use by the consumer, distinction between products on the shelf at the point of sale).

Table 4 summarises the processes identified as the most relevant for changing the nature of foods from raw commodity to derivative in some aggregated food groups, and the new natures (food groups, hierarchy branches) created by these processes in all the aggregated food groups where they apply.

Table 4: Processes identified as the most relevant in some aggregated food groups, and new natures (food groups, hierarchy branches) created by them.

Code ¹	Process	Applied to	Leading to
A0CRH	Seasoning	Meat, milk	→sausages and other charcuterie products, cheese
A07KD	Curing	Meat, milk	
A0C6F	Ripening	Meat, milk	
A07JP	Preserving by salt	Vegetables, meat, fish, seafood	→Salt preserves
A07JQ	Preserving by sugars	Fruit, chestnuts. Candying is included	→Sugar preserves, candied materials and similar
A07KG	Drying (dehydration)	Meat, fish, seafood, milk, egg, vegetables, legumes, fruit. Spices and infusion materials are considered as dried by default	→Dried food
A07JT	Marinating	Meat, fish, seafood	→Marinated meat, fish or seafood
A07KC	Pickling	Vegetables	→Pickled vegetables
A0CQZ	Fermentation	Milk, vegetables, meat	→fermented milk products, fermented vegetables, sausages and other charcuterie products, cheese
A0BYP	Canning / jarring	Vegetable, fruit, meat, fish, seafood	→Preserves
A07JV	Smoking	Meat, fish, seafood	→Smoked meat, fish or seafood
A0C00	Winemaking	Grape and other fruits	→wine and similar
A0C01	Beer production	Malted cereals	→beer
A0C6E	Cheese making	Milk	→cheese
A07MH	Churning	Milk cream	→butter

Code ¹	Process	Applied to	Leading to
A0C02	Oil production	Oil fruits and oilseeds	→oils and fats
A0C03	Grain milling	Cereal grains and similar	→flours, groats etc
A0C0B	Grain milling - starch production	Cereal grains and similar starch sources	→Starch
A0C04	Sugar production	Sugar plants	→Sugar
A0C6N	Pulping / mashing	Fruit, vegetables	→Fruit / vegetable puree
A07GG	Cooking in water	Eggs	→Hardened egg products
A07GR	Frying	Potatoes	→Fries, chips and other potato snacks
A07GX	Baking	Cereal milling products and other Ingredient	→Bakery products
A07GY	Roasting (baking with fat)	Potatoes	→Roasted potatoes
A07HC	Toasting	Coffee beans	→Toasted coffee
A07HJ	Caramelizing / browning	Sugar	→Caramel
A0CRL	Mixing	Creation of compound food	→Recipes (composite food)
A07MA	Filling	Filled branches of unfilled bases	→Filled pasta, filled bakery products
A07LF	Extrusion	Cereals, snack production	→Extruded snacks and breakfast cereals
A07LG	Flaking	Cereals	→Cereal flakes
A07LH	Flattening / rolling	Cereals	→Rolled cereals
A07LL	Puffing / expanding	Cereals	→Puffed cereals
A0C0G	Gelling	Milk Water	→Dessert →Jellies
A07MF	Distillation	Fruits or sugary/starchy sources	→Alcohol and related products
A0EJP	Separation (in liquid phase)	Milk	→Cream
A07MK	Brewing / infusion	Infusion materials, cacao, coffee etc.	→Hot drinks
A07LN	Juicing	Fruit, vegetables	→Fruit and vegetable juices and nectars
A07MJ	Extraction	Meat, Fish, Yeast, Infusion materials	→Extracts
A07KF	Concentration / evaporation	Milk, juices from vegetable and fruit	→Concentrates
A07ML	Crystallisation	Natural sources of purified ingredient	→all 'purified ingredient' like the protein- and carbohydrate-based ones, salt ...

¹FoodEx2 codes for F28-Process descriptors

Example:

Typical examples of derivatives are those listed in Table 5.

Table 5: Non-exhaustive list of food groups considered RPC derivatives in FoodEx2

Code	Food Group	Code	Food Group
A019K	Chili pepper, dried	A022S	Cured seasoned pork meat
A065N	Barley groats	A023N	Pastrami, pork
A002Q	Maize flour	A022L	Animal meat dried
A004D	Chapati flour	A02JS	Cod, dried
A00DR	Rice, popped	A0EYT	Salt-preserved fish
A00ZD	Tomato puree	A02KG	Smoked herring
A00ZK	Sauerkraut	A02NJ	Yoghurt, sheep milk
A011D	Potato flakes	A02QE	Cheese
A013H	Broad beans (dry)	A031Y	Hen egg yolk, dried
A01BD	Canned or jarred lentils	A03MY	Wine, rosé
A017J	Rosemary, dry	A036N	Vegetable fats and oils, edible
A0F0M	Nut/seeds paste/emulsion/mass	A032T	Fructose
A01BN	Peanut butter	A034C	Corn syrup
A03AM	Juice, orange	A011F	Potato starch
A03BY	Juice concentrate, grape	A0EXR	Vitamin B7 (Biotin)
A03GN	Coffee ground, roasted	A0EVV	Omega-3 fatty acids
	Instant coffee powder,		
A03GR	decaffeinated	A048V	Enzymes for food manufacture

In the browser:

Every time there could be a confusion between an RPC and a derivative, the group is divided into non-processed and processed categories (see Figure 12).

If there is a doubt about which term to use, it is recommended to check the category of processed food to see if a *base term* for a derivative exists.

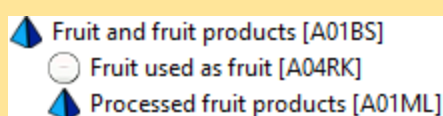


Figure 12: Example of processed and non-processed categories in the catalogue browser

To choose an RPC derivative, it is recommended to look first for the group of the RPC and then to look for a derivative category in this group.

In the practical use of FoodEx2, it happens very often that more than one treatment is applied. It was therefore raised the question of which process-related food group to choose first and which processes to add with facet descriptors:

If for example a fish is smoked, marinated and canned three options would be possible:

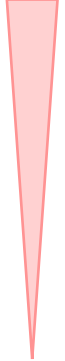
1. *Base term* = marinated fish + facets smoking and canning
2. *Base term* = smoked fish + facets canning and marinating
3. *Base term* = canned fish + facets smoking and marinating

The coding of the same food item might be done therefore in three different ways. In order to avoid the ambiguity, an order of application of the processes (i.e. an order of priority of the related food groups) was defined (see Table 6). The order is substantially arbitrarily defined based on considerations of frequency of reporting, perception of the users and position in the point of sale.

The order is represented in decreasing priority. The first applicable group starting from the top and moving down is used as *base term* and the additional treatments are added as facet descriptor.

Thanks to this approach, the decision on the choice of groups is in most cases automatic.

Table 6: Top-down priority order of use of processed/preserved food groups for the choice of the *base term* in case of multiple treatments

Food group	Underlying Processing	Priority
Puree/textured/structured XXXXX*	Cooking/blanching/separation of parts and mashing or structuring. Eventually associated to canning or packing	
Marinated / pickled / fermented XXXXX	Marinating/pickling/fermenting and eventual canning or packing	
XXXXX in vinegar or brine	Pickling/marinating with acid/salt solution and eventual canning or packing	
XXXXX in alcohol or spirits	Preservation with alcohol and eventual canning or packing	
Salted XXXXX	Preserving by salt	
Candied or sugar-preserved XXXXX	Preserving by sugar	
Dried XXXXX	Drying	
Canned/jarred XXXXX	Packing in glass or can or equivalent closed package and thermal treating as needed for preservation	
Smoked XXXXX	Smoking	
		LOW

*XXXXX replaces the generic name of sections in the classification where the top-down approach applies (e.g. vegetables, meat or fish)

In the browser:

Most of the time, the categories are already ranked in the catalogue browser (see Figure 13).

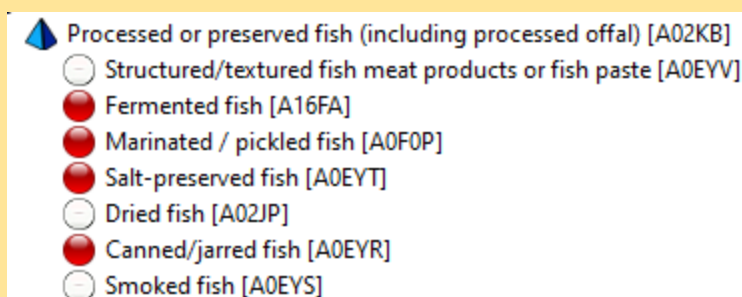


Figure 13: Example of ranked categories in the catalogue browser

Caution must be paid however as some categories are not completely ranked according to the top-down priority of use.

In Figure 14, the ranking is almost right except for the puree/paste category.

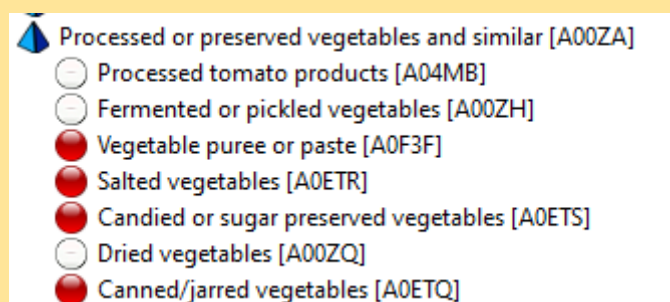


Figure 14: Example of categories not ordered

It is very important to adhere to the order specified in the Table 6 and not to the order of the catalogue browser.

Examples:

Pickled cucumbers in a glass jar go to the group pickled vegetables and the jarring is added as facet.

FoodEx2 code: A00ZJ#F18.A07NN\$F19.A07PF\$F27.A00JM

Translation: Pickled/marinated vegetables, SOURCE-COMMODITIES = Cucumbers, PACKAGING-FORMAT = Jar, PACKAGING-MATERIAL = Glass

Canned haricots don't fall into any of the groups until the canned vegetables group is reached; so, they are coded in the group canned vegetables.

FoodEx2 code: A0ESL

Translation: French beans canned

Marinated fish goes in the marinated group, and if it is also packed or jarred, these are added as facet descriptors;

FoodEx2 code: A0FOP#F18.A07NY

Translation: Marinated/pickled fish, PACKAGING-FORMAT = Envelope/wrap

Smoked sprats in a can go in the group canned fish, with added facet smoking.

FoodEx2 code: A0EYR#F28.A07JV

Translation: Canned/jarred fish, PROCESS = Smoking

2.2.4 The chosen *base term* must be a composite

A composite food *base term* should be considered in all cases **where a product is obtained by combining different Ingredient, no matter the treatments that were applied, using a recipe and a preparation process.**

Composite food items are found in diverse aggregated groups in the FoodEx2 hierarchies, depending on their nature and type of use. The most common food groups for composites are listed in Table 7, which refers to the exposure hierarchy. Similar groupings are also used for other hierarchies.

Table 7: Aggregated food groups of the Exposure hierarchy containing composite food items.

Code	Aggregated food group/Notes
A000J	Grains and grain-based products <i>Bread, pasta, bakery ware, etc</i>
A03BB	Fruit nectars (with a minimum of 25–50 % fruit)
A024F	Sausages
A03TD	Meat and dairy imitates
A03DJ	Water and water-based beverages <i>With the exclusion of water, the others are mainly composite</i>
A03JZ	Hot drinks and similar (coffee, cocoa, tea and herbal drinks)
A03NS	Liqueurs
A04PE	Confectionery including chocolate
A03PV	Food products for young populations <i>With the exception of few derivatives</i>
A03RR	Food for particular diets
A03VA	Composite dishes
A042N	Seasoning, sauces and condiments

The list is not exhaustive, but the most commonly addressed groups for data reporting are listed.

In the case of a composite food item, the choice of the *base term* is driven by its main ingredient. The *base term* is chosen according to the most distinctive aspect of the food item.

If the choice of a group for a composite food is easy in all cases when one of the ingredient is clearly predominant or significant - in practice, a large proportion of recipes have many important ingredients in quantitative balance.

When the major ingredient of a composite is not clearly identified, a top-down priority in the choice of the *base term* was built into the food list following the criteria listed in Table 8. The criterion chosen is only pragmatic and based on nutrient source and the intention of the consumer choosing the dish. Once a *base term* has been chosen for the composite food, the other important ingredient can be added as characterising ingredient (with the F04-*Ingredient* facet).

Table 8: Top-down priority order of use of food groups for composite dishes (and similar cases).

Food group	Assumed intention while choosing the food	Priority
Meat based...	Source of protein	HIGH 
Fish based...	Alternative source of protein	
Cheese or dairy based...	Source of protein and fat	
Egg based...	Source of protein and fat	
Legume based...	Source of protein and carbohydrates / energy	
Potato based...	Source of carbohydrates / energy	
Cereal based...	Source of carbohydrates / energy	
Fruit based...	Source of micronutrients	LOW
Vegetable based...	Source of micronutrients	

The predefined order facilitates the choice of the base term in case of multiple characterising ingredients without a clear predominance of one of them.

Examples:

A **meat stew with potato**: meat-based dish unless the potatoes become really quantitatively predominant.

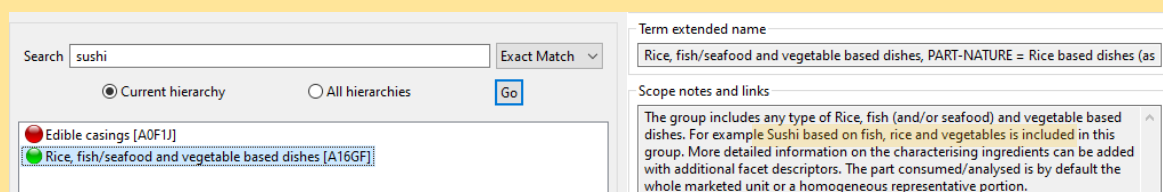
A portion of **chili con carne** is much depending on the recipe but the legumes may be in most cases the major ingredient.

However, even with this “top-down priority” method, several interpretations remain possible for categorising numerous “composite dish” types of foods.

In the Catalogue Browser:

The first thing to do when coding a composite dish is to search it into the research panel. Even if the dish may not have a specific *base term*, the catalogue may propose different options.

In the example below (see Figure 15), there is no *base term* for sushi but the catalogue proposes a category for the *base term* and even suggests an ingredient for the seaweed surrounding the rice (“edible casings” [A0F1]), to further describe the food item.

**Figure 15:** Example of a search for a *base term* in the catalogue

To choose a base term, one should not refer to the order in the catalogue browser but to the order presented in Table 8. The classification in the catalogue browser is actually based on the creation date of the terms, not on the priority order for selecting a base term.

2.2.5 Difference between a composite and a mixed food product

Some « mixed food products » are simply a mix of unprocessed food items. Following the logic of the system, the presence of multiple ingredients should define a composite food. But in this case, they are not generated by recipes like a classic composite food. Each component of such products is actually a raw commodity or a derivative and is not processed in a way that significantly modifies it or somehow chemically combines it with other components; such products are simply mixtures.

Even though these food items are, strictly speaking, composites, in some cases, such as for exposure assessments, it is more convenient to code them as raw commodities or derivatives of raw commodities. The main reason for this preference is that composite food groups are more heterogeneous and, in general, more difficult to analyse in risk assessments. So, unless they already existed in the system, specific composite groups were not created for those kinds of mixed food products.

Examples:

Mixed milks (from different animal species), mixed lettuce, mixed beans, mixed flours, mixed grains, mixed breakfast cereals, mixed nuts

If the process simply involves mixing the ingredients, without additional significant processes, then a decision must be made, according to the type of mixture (see 3.2 for more detailed information):

- **A mixture of raw commodities/ingredients of the same nature** or substantially of the same nature should be coded as a raw commodity or derivative for the *base term*

Examples:

Mixed milks, mixed leafy vegetables, mixed nuts and mixed minced meat or mixed leafy vegetables with some carrots, mixed nuts with some raisins and mixed nuts with some oilseeds.

- **A mixture of raw commodities/derivatives of different natures and in balanced amounts**, without a clear predominance of commodities with a specific nature (corresponding to a generic term), should be coded as a composite. In these cases, the best starting point should be found on a case-by-case basis. Criteria could be related to the type of use (salad, soup or snack), then the most appropriate generic starting point can be chosen.

Examples:

Mixes for minestrone and ready mixes for elaborated salads

2.2.6 Level of details accepted

Once the type of the food is found, finding the right *base term* is simple.

However, attention must be paid to the level of detail of the *base term*:

- **Never use Hierarchy terms** (blue pyramids in the browser tool )
- As much as possible, give priority to **Core terms** (red ) and **Extended core terms** (green ) if relevant.

When more information is not available, generic terms (nature and source unknown) (white ) and non-specific terms (nature known but unspecified origin) (yellow ) can also be used for reporting data.

The **Generic terms allow** a non-hierarchy term to be chosen in two particular cases:

- When the details (e.g. the source) are not known by the coder and only the observation of the food sample may provide an indication (the *base term* is unspecified);
- When the details are available to the coder, but the detailed term is missing in the food list (the required *base term* would be 'other' than the ones available in the list).

The **Non-specific terms** are found at many points of the food list. They principally cover the following two possibilities:

- Groups of 'others' (used in data reporting or even in data collection for minor food items in a broader group);
- Intermediate groups between generic terms and core terms - they provide a better way of splitting groups while navigating the system or in data analysis.

Simply put, the following order of priority should be used for the choice of a *base term*:

Extended term > Core term > Non-specific term > Generic term

2.3 AFTER THE *BASE TERM*, WHAT FACETS TO USE?

2.3.1 Implicit facets

Never report implicit facets (cf. 1.2) as they can easily be added at the time of data analysis if needed. However, they can guide the choice of facet to add in two ways:

- If the chosen term is not precise enough in the origin of the food, the implicit facet can orient the facets to add. See 2.3.5
- If the precise term does not exist for our item but a similar one exists for another, its implicit facet can orient the choice of facets. See 2.3.6.2

2.3.2 Cardinality and applicability of the facets

FoodEx2 includes many facets. Not all of them are equally applicable to all terms and the use of some terms is only appropriate for a subset of possible *base terms*. For example, the F31-*Animal-age-classes* facet (e.g. Adult mammal, Hatching egg, Day-old chick) or the F30-*Reproductive-level* facet (e.g. Grandparent breeding animal, Parent breeding animal, Animals not for breeding) can only be used for animals and do not make sense for plants.

Cardinality means the number of descriptors of the same facet that can be applied to a single *base term*. For both applicability and cardinality of the facets, general rules are defined, and some exceptions are foreseen in particular approaches to coding.

- Some facets can only support one descriptor (see Table 9):

Table 9: Facets which descriptors can only be applied once to a *base term*

Facet	Cardinality	Applicability
F01 - Source	Single	Normally applicable to raw commodities. In rare cases to derivatives of raw commodities or composite food to better specify single generic source commodity
F02 - Part-nature	Single	All
F03 - Physical-state	Single	All
F07 - Fat-content	Single	All
F11 - Alcohol-content	Single	Alcoholic beverages
F22 - Preparation-production-place	Single	All
F24 - Intended-use	Single	All
F26 - Generic-term	Single	All
F27 - Source-commodities	Single (<i>Repeatable for mixed raw commodities or derivatives only</i>)	Normally to derivatives of raw commodities. Also to raw commodities in the mixed raw commodities approach.
F30 - Reproductive-level	Single	Animals
F31 - Animal-age-class	Repeatable	Animals and their primary products
F32 - Gender	Single	Animals and their primary products
F34 - Host-sampled	Single	Animal

- Some other facets can support multiple descriptors (see Table 10):

Table 10: Facets which descriptors can be applied multiple time to a *base term*

Facet	Cardinality	Applicability
F04 - Ingredients	Repeatable	Normally applicable to composite. In the minor ingredient approach, also to raw commodities or derivatives of raw commodities
F06 - Surrounding-medium	Repeatable	All, but in particular to canned products or products packed in a close packaging
F08 - Sweetening-agent	Repeatable	All
F09 - Fortification-agent	Repeatable	All
F10 - Qualitative-info	Repeatable	All
F12 - Dough-Mass	Repeatable	Bread and Fine bakery wares
F17 - Extent-of-cooking	Repeatable	All (in particular raw commodities and composite)
F18 - Packaging-format	Repeatable	All
F19 - Packaging-material	Repeatable	All
F20 - Part-consumed-analysed	Repeatable	All
F21 - Production-method	Repeatable	Animals, plants and their primary products
F23 - Target-consumer	Repeatable	All
F25 - Risky-Ingredients	Repeatable	Composite
F28 - Process	Repeatable	All
F29 - Purpose-of-raising	Repeatable	Animals
F33 - Legislative-classes	Single for each sub-list; Repeatable among sub-lists	All

2.3.3 How many facets should be used?

The number of added facet descriptors is in principle not limited, but for practical purposes, only the facets allowing better defining a sub-group inside a food group should be added.

Examples:

Cooked carrot, sliced:

A00QH#F28.A07GL\$F28.A07KV

Translation: Carrots, PROCESS = Boiling, PROCESS = Slicing

Apple, organic:

A01DJ#F21.A07SE

Translation: Apples, PRODUCTION-METHOD = Organic production

2.3.4 General and specialised facets

The general facets can, in principle, be reported in different domains. However, some of them do provide quite specific information and their use is limited:

F07, F11 - Fat or Alcohol -content, F09-Fortification-agent, F21-Production-method, F17-Extent-of-cooking, F06-Surrounding-medium, F12-Dough-mass, F22-Preparation-production-place, F23-Target-consumer and F03-Physical-state

The specialised facets (*F24-Intended-use, F25-Risky-Ingredients, F29-Purpose-of-raising, F30-Reproductive-level, F31-Animal-age-class, F32-Gender and F33-Legislative-classes*) are mainly expected to be used within their specific domains of applicability; the remaining domains would not normally use them.

The potential high number of facets should not scare the coder, because the focus is on only a few of them, on a case-by-case basis. Experience shows that the facet descriptors added are normally few and from a limited number of facets.

The most common use of the facets available in FoodEx2 is summarised in Table 11.

Table 11: FoodEx2 facets and common cases for their use.

Facet	Cases of use
F01 - Source	Implicit. Sometimes added
F02 - Part-nature	Implicit. Sometimes added
F03 - Physical-state	To be used rarely to distinguish between two forms of the same food
F04 - Ingredients	Very important , but the focus is on the characterising ingredient
F06 - Surrounding-medium	Very important but only for food in packages such as cans, jars or plastic containers with the presence of a surrounding medium
F07 - Fat-content	Only used when this information better characterises the food
F08 - Sweetening-agent	Important for non-sugar-sweetened sweet products
F09 - Fortification-agent	Important for fortified food products
F10 - Qualitative-info	Very important for different specific characteristics. Different sections inside the facet address separate aspects
F11 - Alcohol-content	Only relevant in the category of alcoholic beverages, when the alcohol content is not obvious (in a reasonable range)
F12 - Dough-Mass	Only relevant for bakery products, when not obvious (to better characterise the food)
F17 - Extent-of-cooking	Important for products such as grilled meat or vegetables, etc.
F18 - Packaging-format	Importance defined by the purpose of the data collection
F19 - Packaging-material	Importance defined by the purpose of the data collection
F20 - Part-consumed-analysed	Important in some cases, to better specify the sub-part of a food that is referred to
F21 - Production-method	Important in some cases (such as organic farming, aquaculture, etc.)
F22 - Preparation-production-place	Importance defined by the purpose of the data collection
F23 - Target-consumer	Only to be used when not obvious, or when this information makes the difference
F24 - Intended-use	<i>Used only in the microbiological domain</i>
F25 - Risky-Ingredients	<i>Used only in the microbiological domain</i>
F26 - Generic-term	Very important in all cases when a generic term is used.

Facet	Cases of use
F27 - Source-commodities	Implicit. In some cases added
F28 - Process	Very important. Partly implicit, partly to be added. Only the treatments 'making the difference' have to be mentioned
F29 - Purpose-of-raising	<i>Only relevant in the animals domain</i>
F30 - Reproductive-level	<i>Only relevant in the animals domain</i>
F31 - Animal-age-class	<i>Only relevant in the animals domain</i>
F32 - Gender	<i>Only relevant in the animals domain</i>
F33 - Legislative-classes	<i>Very important for just some legislation-oriented data collections, such as food additives data or microbiological data</i>
F34 - Host-sampled	<i>Only relevant in the animals domain</i>

2.3.5 Which facet should be used to describe the origin of the base term?

The three following facets: F01 – *Source*, F27 – *Source-commodities* and F04 – *Ingredients*, include more or less the same descriptors, answering to the question:

What is the origin of this food item? What is it made from?

The choice of the facet is determined by the type of food group to which the *base term* chosen belongs (RPC, RPC derivative or composite). The main rule to remember is summarised in the table below (Table 12), more described in the paragraphs below:

Table 12: Facets describing the origin of the different types of food groups

Type of food groups	Facet describing the origin (what is it prepared from?)	Notes and examples
Raw commodities (RPC)	F01-Source	Raw commodities, such as roots and leaves, are derived from a live plant source; for example, beetroot is obtained from the <i>Beta vulgaris</i> plant
Derivatives of raw commodities (RPC derivatives)	F27-Source-commodities	Derivatives, e.g. flour and oil, are produced from a source that is normally a raw commodity (source raw commodity); for example, wheat flour is produced from wheat grain, while sunflower oil is produced from sunflower seeds
Mixed food product		<i>Mixed food products can be a mixture of RPC or derivatives of the same nature, e.g. mixed nuts, then described with the Source-commodity facet</i>
Composite food	F04-Ingredient	Inside a defined composite group, such as salads or risotto, the characterising ingredient define different sub-groups; for example, seafood risotto is characterised by the ingredient 'seafood'
Minor Ingredients		<i>Minor ingredients added to RPC or derivatives can also be described with the ingredients facet, e.g. for the chocolate in chocolate-coated nuts</i>

All this information is summarised in the ANNEX 4: FACETS SUMMARY → F01, F27 OR F04 FACETS?, page 95.

Based on the chosen base-term, these facets often already have **implicit descriptors**. When it's necessary to specify the base-term with facets, it's easier to start with the implicit facets already associated with the base-term.

Example: Veal brain

In FoodEx2, there is a term A020R - Bovine other slaughtering products. This base-term has 3 implicit facets:

- F27. A020R "Bovine other slaughtering products"
- F02.A0BYL "Other slaughtering products (as part-nature)", which can be replaced with the more specific facet F02.A06AM "PART-NATURE = Brain (as part-nature)"
- F01.A0F1V "Bovinae (bovines = Tribe bovini) (as animal)", which can be replaced with the more specific facet F01.A0C9E "SOURCE = Calves (under 1 year) (as animal)" or F01.A0C9M "SOURCE = Veal calves (under 1 year) (as animal)"

Another term could have been used, "Bovine edible offal, non-muscle, other than liver and kidney" [A01ZM] which has, amongst its implicit facets, F02.A0BYK "Other organs (non-muscle) (as part-nature)". However, in the F02-*Part-nature* facet, this general descriptor is not parent of the descriptor "Brain (as part-nature)" [A06AM]. It signals that the term A01ZM is not a good base-term for our code.

2.3.5.1 The F01-Source facet

The **F01-Source facet** is in many cases **implicit**, so it does not have to be reported by the coder. Even so, in some cases, this facet can provide more information than the *base term* available.

When a descriptor of the F01-Source facet is implicitly included in an RPC *base term*, **and if** the information of the food sample allows specifying a more detailed source, **then** the F01-Source facet should be added by the coder as a 'restriction', i.e. by choosing a more detailed sub-group of the existing source (a children code of the implicit code already existing).

In this case, the starting code should be as follow:

RPC *base term* + Facet F01-Source

Example:

To code a specific type of sturgeon, such as the Adriatic sturgeon, the system only has "Sturgeon" [A029E], including the implicit facet F01.A04XV "SOURCE = Sturgeons (generic) (as animal)". However, the descriptor F01.A0884 "SOURCE = Adriatic sturgeon (as animal)" exists.

In this case, the final code would be "A029E#F01.A0884"
Sturgeon, SOURCE = Adriatic sturgeon (as animal)

In the same way, the origin of an RPC derivative (or a composite) is reported by a single descriptor of the F27-Source-commodities facet (or the F04-Ingredient facet). **If** the exact original source is missing from the list of terms in these two facets (where only a generic term is available), **then** it can be clarified using the F01-Source facet. Since a single F27-Source-commodities facet (or a single F04-Ingredient facet) is reported, the eventual addition of a F01-Source facet is not ambiguous and may be assumed, by default, as referring to it (like a 'facet of the facet'). So, in this very particular case, the non-standard use of the F01-Source facet may be tolerated and even prove to be useful.

In this case, the starting code should be as follow:

RPC derivative + Facet F01-Source + Facet F27-Source commodities

Example:

If the Adriatic sturgeon of the previous example had been smoked, the correct code would become "A0EYS#F27.A029E" (Smoked fish, SOURCE-COMMODITIES = Sturgeon) and the information on the actual source of the sturgeon meat would be lost. However, if we also add the F01-Source facet descriptor (otherwise not allowed in a derivative), the code becomes 'A0EYS#F01.A0884\$F27.A029E' (Smoked fish, SOURCE = Adriatic sturgeon (as animal), SOURCE-COMMODITIES = Sturgeon). This is possible because we have a single source raw commodity and we can define that the presence 'in derogation' of the source facet refers in this case to the only present source raw commodity.

2.3.5.2 The F27-Source-commodities facet

Like the F01-Source facet, the **F27-Source-commodities facet** is in many cases **implicit**, so it does not need to be reported by the coder.

For derivatives, attention has to be paid, because there is a natural 'cascade' mechanism: derivatives are produced from a raw commodity and the raw commodity itself is produced from a live plant.

Example:

Rice flour is produced from the rice grain by applying a milling process and the rice grain is produced from the rice plant (*Oryza sativa*). The cascade mechanism is always traceable within the logic of the system, but the relationship between facets and the food group must be respected. Therefore, for rice flour, the direct source is the rice grain, not the *Oryza sativa* plant.

The same idea, with a further step, also applies to composite food. White bread is made from wheat flour, which is produced from wheat grains which are obtained from the wheat plant (*Triticum aestivum*). In this case, the correct facet for describing the origin is the ingredient and, therefore, the right descriptor is the wheat flour, not the wheat grain (as source raw commodity) or the *Triticum aestivum* plant (as source).

However, in some cases, the **F27-Source-commodities facet** can be used to further detail the composition of certain food such as mixed products of the same nature (whether they are RPC or derivatives) or to further describe the source of an RPC derivative, already described with an implicit facet.

Example:

To code flour obtained from glutinous rice, since only the more generic term "Rice flour" [A003F] exists (having as an implicit facet F01.A001D "SOURCE = rice grain"), coding should use this term but specify, by restriction, the more precise "Rice grain, glutinous" [A0F6M] a F07-Source-commodities facet descriptor.

The final code becomes: A003F#F27.A0F6M

2.3.5.3 The F04-Ingredient facet

The **F04-Ingredient facet** is normally used to report the **characterising ingredient(s) in a composite food**.

This facet is not intended to provide the full qualitative recipe by mentioning all the ingredients. Its purpose is to specify those ingredients that, inside a composite food group, make this food product different from the others. Normally only one or few descriptors for the F04-Ingredient facet are expected in a FoodEx2 code.

In the case of ingredients used to provide a specific taste, the ingredient should be reported as raw commodity to simplify and standardise the reporting.

Example:

To code a risotto with asparagus, the ingredient F04.A00RT "INGREDIENT = asparagus" should be used even if it was added in dried form or as preparation. The final code becomes A041F#F04.A00RT (Risotto, INGREDIENT = Asparagus).

It is important to keep in mind that even if other ingredients, like fats, milk components etc. may be present in the product, they must not be mentioned in the FoodEx2 code since only the characterising ingredient are relevant. The eventual management of recipes is not foreseen inside FoodEx2; recipes should be collected in an external database (that may be coded in FoodEx2), not recorded with the F04-Ingredient facet.

In addition, the F04-Ingredient facet can also be used to **describe the presence of minor ingredient** as flavouring or coating (or for decoration purposes) **in raw commodities or derivatives of raw commodities**.

Example:

In case of candied citrus peel, chocolate coated, the *base term* is "Candied fruits" [A01PS]. The descriptor F27.A01QE "SOURCE-COMMODITIES = Citrus fruit peel" is added as origin; finally, the chocolate is added as minor ingredient and coating is added as process. The final code becomes A01PS#F04.A034G\$F27.A01QE\$F28.A07HP (Candied fruits, INGREDIENT = Bitter chocolate, SOURCE-COMMODITIES = Citrus fruit peel, PROCESS = Chocolate coating)

A list of examples of cases where this 'minor ingredient' approach would apply is reported Table 13:

Table 13: List (non-exhaustive) of food items where the 'minor ingredient' approach applies

Food items or groups thereof	Notes
Salted nuts and seeds	
Sweetened nuts and seeds	
Chocolate-coated nuts, seeds, fruits	
Meat prepared with some oil or spices	For marinated products, a specific derivative group exists
Fish or seafood prepared with some oil or spices	For marinated products, a specific derivative group exists
Cheese (fresh or ripened) with herbs or spices	
Oil with herbs or spices	
Butter with herbs	
Liqueurs or vinegar with herbs in the bottle	
Sugar-coated nuts, seeds and similar	
Nut mixtures with some seeds or raisins	For heterogeneous (different natures) mixes, when a homogeneous (same nature) food group predominates, the approach applies
Flour with seeds	
Pieces of meat with a bacon or suet coating	
Flavoured bottled waters	

All ingredients contributing to the characterising flavour should be mentioned in the FoodEx2 code.

Flavoured products may be ambiguous for the coder, because they may be flavoured with ingredient (e.g. strawberries), with intrinsic flavour not falling under the scope of Regulation (EC) No 1334/2008 or with flavourings (e.g. strawberry flavour) falling under the scope of Regulation (EC) No 1334/2008, or with both. If limited information is available, at least the ingredient (e.g. strawberries) should be mentioned.

Single flavouring ingredient should only appear in those products (e.g. some soft drinks and some yoghurt types) where only regulated flavourings are used and there is no other food ingredient (fruit, spices, etc.) contributing to the flavour note.

2.3.5.4 Decision charts for facets describing the origin

Here are the decision charts designed to help choose the appropriate facets based on the type of the *base term* (see Figure 16):

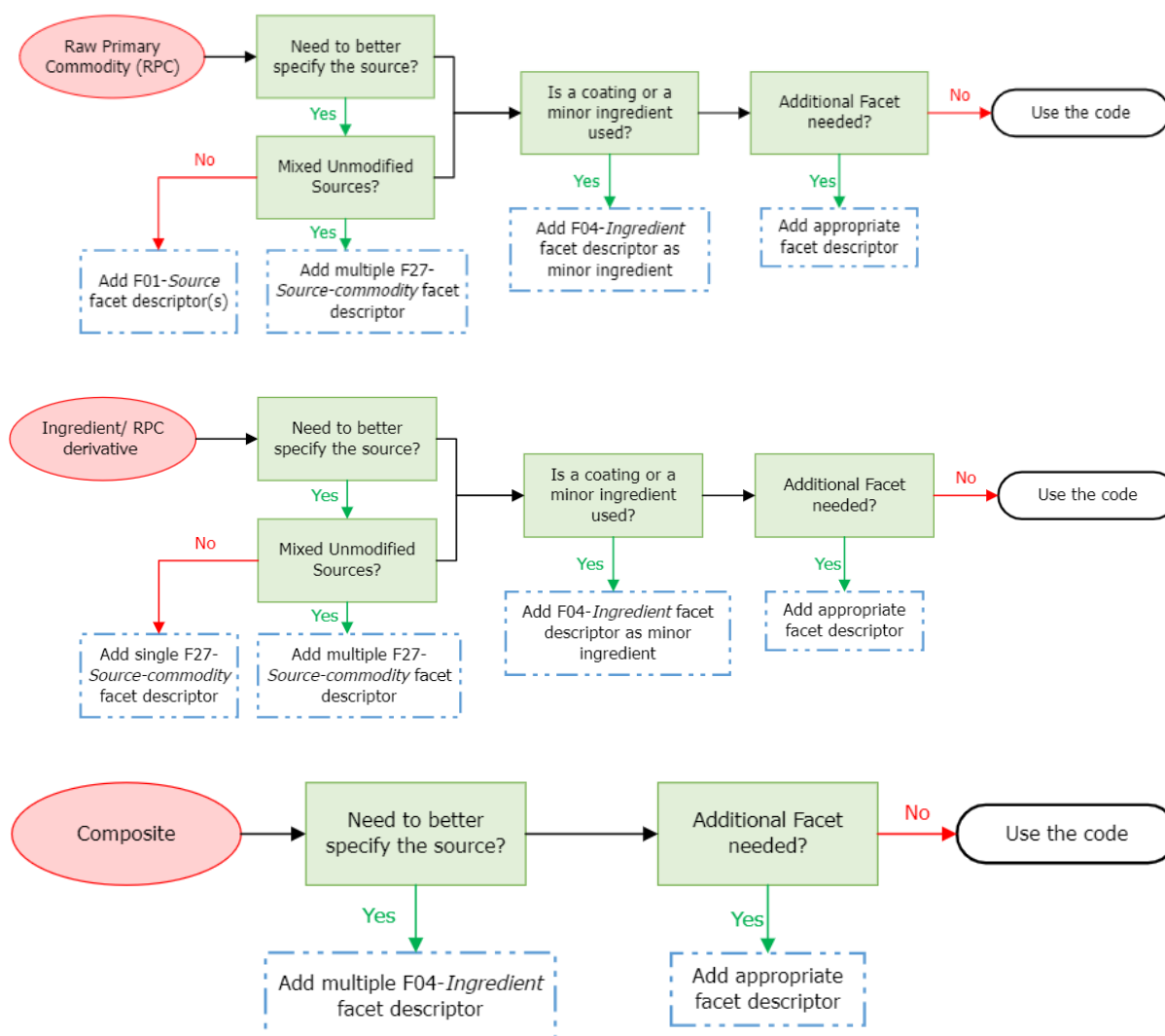


Figure 16: Decision chart for facets describing the origin

2.3.6 The F28-Process facet

Sometimes, similar descriptors can be found in the F28-Process facet and the F10-Qualitative-info. In such cases, the descriptor of the F10-Qualitative-info must be used.

Example:

To describe a product light in lactose, there is the descriptor F10.A0B8N "QUALITATIVE-INFO = Low / reduced lactose" and the descriptor F28.A07MM "PROCESS = Lactose reduction". In this case, the descriptor to use should be the Qualitative-info descriptor "Low / reduced lactose".

2.3.6.1 If the base term is an RPC

Only treatments not creating a new nature can be applied (as added facets) to raw commodities *base terms*. A list of process descriptors that are normally applied directly to the raw commodity without referring to a new derivative nature is provided in Table 14. The list intends to cover the most common ones.

In rare cases, a specific group applying one of these processes exists (mostly due to legacy food groups not removed from the system). The generic process descriptors of the Process facet, "Processed", "Preserved" and "Semi-preserved", do not refer to a specific process, therefore they are not considered.

Table 14: Examples of processes normally directly applied to the raw commodity; specific derivative groups have generally not been created for them.

Process facet descriptors	Exceptions (when a derivative group exist)
▼  Physical division / dimension reduction [A07KS]  Portioning [A07KT]  Slicing [A07KV]  Dicing and stripe-cutting [A07KX]  Mincing / chopping / cutting [A07KY]  Grating [A07KZ]  Grinding / milling / crushing [A07LA]  Pulping / mashing [A0C6N]	Some groups exist: • Minced meat • Milling associated to separation processes creates the grain derivatives, however specific Grain milling descriptors are applicable instead • Mashing, associated with separating parts and eventual cooking leads to puree groups existing in some groups of derivatives of plant origin.
 Preservation treatments not changing nature [A0BYG]  Physical decontamination treatment [A07JB]  Aspiration [A0BZG]  Desliming [A0BZL]  Cleaning [A07JE]  Brushing [A07JF]  Washing [A07JG]  Centrifugal cleaning [A07JH]  High pressure treating (pascalisation) [A07JC]  Micro-filtering [A07JD]  Irradiation [A07JJ]  Surface treatment with sanitizing agents [A0CSQ]  Lowering temperature [A07KN]  Chilling [A07KP]  Freezing [A07KQ]  Iqf freezing (flash/blast freezing) [A07KR]	

Process facet descriptors	Exceptions (when a derivative group exist)
▲ Thermal treatment (heating for preservation) [A07HR] ● Low pasteurisation (thermisation) [A07HT] ● Pasteurisation [A07HV] ● High pasteurisation (extending shelf life) [A07HX] ● UHT [A07HY] ● Static sterilisation (in batch or package) [A07HZ] ● Hot-filling [A07JA]	
✓ ▲ Protective packing [A0BYM] ● Aseptic filling [A0BYN] ● Canning / jarring [A0BYD] ● Vacuum-packing [A07JK]	
▲ Compounding, assembling, coating or filling [A07LZ] ● Blending [A0GRJ] ● Mixing [A0GRL] ● Filling [A07MA] ● Breeding [A07HK] ● Battering [A07HL] ● Glazing / icing [A07HM] ● Sugar coating [A07HN] ● Chocolate coating [A07HP] ● Nuts coating [A07HQ]	
● Cooking and similar thermal preparation processes [A0BA1] ● Blanching [A07GF] ● Cooking in water [A07GG] ● Stewing [A07GM] ● Bain-marie cooking (in water bath) [A07GN] ● Steaming [A07GP] ▲ ● Frying [A07GR] ● Air-frying [A18TD] ● Baking [A07GX] ▲ ● Roasting [A07GY] ● Grilling/griddling (high temperature cooking) [A0EJY] ● Microwave-cooking [A07HB] ● Infra-red micronisation [A0CRA] ● Toasting / coffee roasting [A07HC] ● Reheating [A07HD] ▲ ● Caramelization / browning [A07HJ]	• Frying/roasting for potatoes creates new derivatives • Cooking eggs creates new derivatives • Caramelisation / browning creates derivatives e.g. in sugar and milk

Crossed out group have derivative term created in the classification

2.3.6.2 If the base term is an RPC derivative or composite

In most cases, the choice of a derivative or composite group already implicitly refers to a treatment with one (or several) descriptors of the F28-Process facet added as implicit facets. Process should not be mentioned in the code. However, it is important to check if the process was indicated as implicit facets. Otherwise, it has to be added as explicit facet (as in the code of the food item) and if possible, signal the EFSA about the missing implicit facets.

Implicit facets can be found in the catalogue browser, in the information panel, as shown before Figure 10, page 23.

Example: Cereal flour

The term "Cereal and cereal-like flours" [A04KS], has the following implicit facets:

- F02.A067Z - PART-NATURE = Flour/meal or finely ground powder (as part-nature)
- F27.A000L - SOURCE-COMMODITIES = Cereal grains (and cereal-like grains)
- F28.A0C0A - PROCESS = Grain milling - flours production
- F03.A06JD - PHYSICAL-STATE = Powder

The F28.A0C0A indicate the transformation process and the F03.A06JD the physical state of the item. There is no need to add another PROCESS facet descriptor.

Additional processes can be added if needed. Looking at the implicit facets of terms is a good way to find which facets to add.

Ex: Chestnut flour

The base term to use is: “Dried nuts/seeds and related flours and powders” [A0F0N], which is not specific enough to characterise a flour as the implicit facets implies:

- F02.A0ELZ: PART-NATURE = Dried nuts and related flours and powders (as part-nature)
- F28.A07KG: PROCESS = Drying (dehydration),
- F33.A0C19: LEGISLATIVE-CLASSES = FA-15.2 Processed nuts

However, there is a term “Pulses flour” [A185Q] whose implicit facets are, amongst others:

- F28.A07LA: PROCESS = Grinding / milling / crushing
- F03.A07Y3: PHYSICAL-STATE = Coarse powder

Taking those facets as reference,

The final code can be: A0F0N#F03.A07Y2\$F27.A014J\$F28.A07LA

Translation: Dried nuts/seeds and related flours and powders, PHYSICAL-STATE = Fine powder, SOURCE-COMMODITIES = Chestnuts, PROCESS = Grinding / milling / crushing

2.3.7 The F26-Generic facet

This facet contains two descriptors:

F26.A07XE “GENERIC-TERM = Other”:

- If the FoodEx2 code created is insufficient to fully describe the sample, this facet suggests that there are insufficient descriptors in FoodEx2 to adequately represent it. (= “I have too much information for FoodEx2”)
- This information is readily available and facilitates the proposal of new terms during system updates. Therefore, it is strongly recommended to include this facet only when necessary.

F26.A07XD “GENERIC-TERM = Unspecified”:

- If you don’t have enough information from your sample to describe entirely into FoodEx2 (= “I don’t have enough information”)
- To differentiate a plain food, i.e. plain biscuit, from an unspecified food, i.e. biscuit unspecified.

3 PARTICULAR RULES OF CODIFICATION

This next section presents some general rules that applies for particular codification issues. An annex is also provided containing all the terms that may have been problematic in previous codifications in FoodEx2 and the code chosen to help harmonise our practices, see ANNEX 1: PRACTICAL EXAMPLES OF FOODEX2 CODES, page 73.

3.1 WHAT TO DO WHEN THE REQUIRED *BASE TERM* IS MISSING?

The general rule is to use the nearest upper-level group specifying the additional information with facets if available.

The facet F26.A07XE “GENERIC-TERM = Other” should also be added to communicate that the detailed term was not found in FoodEx2 and the description was built with facets.

Different rules apply depending on the type of information available, as described below:

3.1.1 The Raw Primary Commodity base term is missing

If a Raw Primary Commodity term is missing, the corresponding generic term should be present. The list of live animals or plants (those used as source of the raw commodities) is in general more comprehensive than the list of commodities so it often happens that a suitable source to further specify a generic food group is available. In this case the code is built using the **generic food group** as *base term* and adding the suitable source with the **F01-Source facet**. The starting code is then:

RPC generic term
+ Facet F01-Source
+ descriptor F26.A07XE « *GENERIC-TERM = Other* »

And this can be further described with other facets.

If the facet F01-Source is also missing, then the only possibility is to stay general in the FoodEx2 code. The complete information is recorded in the wording of the food item.

Examples (in the Reporting hierarchy):

Kangaroo fresh fat tissue: The *base term* doesn't exist in the hierarchy but the descriptor F01.A0F2G “SOURCE = Kangaroo, as live animal” does. The nearest generic food group is “Mammals fat tissue” [A0F3G] and can be used as *base term*.

The code is: **A0F3G#F01.A0F2G\$F26.A07XE**

(Mammals fat tissue, SOURCE = Kangaroo (as animal), *GENERIC-TERM* = Other)

Eagle eggs: The generic food group “Whole eggs” [A031F] exists, but the source descriptor “Eagle” doesn't. In this case the code must be **A031F#F26.A07XE** (Whole eggs, *GENERIC-TERM* = Other) and the source=eagle will not appear in the code

The method in this case is described below, see Figure 17.

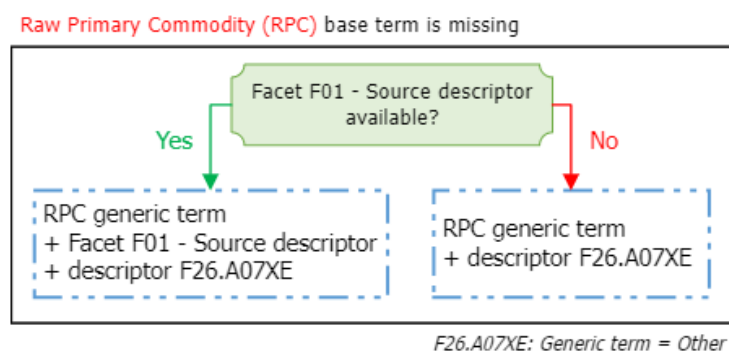


Figure 17: What if the RPC *base term* is missing?

3.1.2 The RPC derivative *base term* is missing

If a **derivative of raw commodity term is missing**, then a more generic term has to be used. In this case, the code is built by using the generic food group as the *base term* followed by the suitable descriptor with the F27-*Source-commodities* facet.

The starting code is then as follows:

RPC generic term
+ Facet *Source-commodities* descriptor
+ descriptor F26.A07XE « *GENERIC-TERM* = Other »

And this can be further described with other facets.

Examples (in the Reporting hierarchy):

Quinoa flour: The group doesn't exist as such in the food list. However, the descriptor F17.A000R "SOURCE-COMMODITIES = Quinoa grain" exists. The nearest generic food group is "Cereal and cereal-like flours" [A04KS].

The code is therefore: **A04KS#F26.A07XE\$F27.A000R**

(Cereal and cereal-like flours, GENERIC-TERM = Other, SOURCE-COMMODITIES = Quinoa grain).

Maize groats: Among the pre-defined groats, maize is not present. Coding this product is possible, but should start from the generic term "groats" [A0BY1] and the source should be specified with the F27-*Source-commodities* facet.

The final code, in this case, would be: **A0BY1#F26.A07XE\$F27.A000T**

(Groats, GENERIC-TERM = Other, SOURCE-COMMODITIES = Maize grain)

If there is no suitable descriptor in the F27-*Source-commodities* facet but the original source of the derivative is available in the **F01-Source facet**, then adding this descriptor to a derivative is allowed as it would provide more information than simple text would. This particular use of the F01-*Source* facet, to describe a derivative missing in the food list, is suggested because it does not introduce ambiguity and 'explicitly' provides information that is implicit in all other cases. Please note that this is one of **very few cases** where the use of the source facet for food groups not representing raw commodities may be accepted whereas it goes against the cascade mechanism explained in 2.3.5.2, page 41.

Examples (in the Reporting hierarchy):

Dried kangaroo meat: This derivative does not exist in the food list; even the F27-*Source-commodity* descriptor (kangaroo meat) is missing. However, the live source animal, F01.A0F2G "SOURCE = kangaroo, as live animal", is available. In this case, the nearest generic food group is "mammals or birds dried meat" [A04MP].

The code is therefore: **A04MP#F01.A0F2G\$F26.A07XE**

(Mammals or birds dried meat, SOURCE = Kangaroo (as animal), GENERIC-TERM = Other)

The source (i.e. from kangaroo (meat)) is understandable from the code.

If there is no suitable **F01-Source descriptor**, then the information will not be translated in the FoodEx2 code but it is understandable from the food wording.

Examples (in the Reporting hierarchy):

Pelican egg yolk: The generic food group "egg yolk" [A0F6F] exists, but there are no "pelican egg" or "pelican" descriptors.

The appropriate code would be "**A0F6F#F26.A07XE**" (Egg yolk, GENERIC-TERM = Other) and the source = pelican egg will not be present in the code.

The method in this case is described below, see Figure 18.

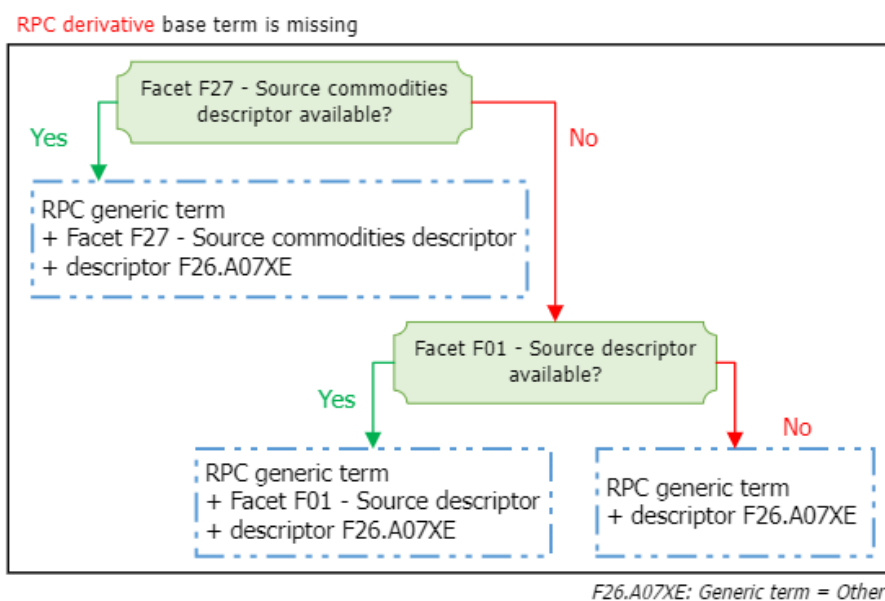


Figure 18: What if the RPC derivative *base term* is missing?

3.1.3 The composite *base term* is missing

Composite food groups are present in different areas of the classification tree. The choice of the correct 'branch' of the tree and, within this branch, the choice of a *base term* (non-hierarchy term) that is conceptually similar to the product being coded is done based on a criterion of similarity of use, position in the meal and expected major ingredients.

If the precise composite term corresponding to the food does not exist, a similar term must be chosen, the characterising ingredients should be provided and the facet F26.A07XE "GENERIC-TERM = other" should also be added.

Example:

Salmon sushi: As mentioned previously, there is no composite *base term* "sushi". The nearest term would be « Rice, fish/seafood and vegetable-based dishes » [A16GF].

The code would be **A16GF#F04.A0C75\$F26.A07XE**

Rice, fish/seafood and vegetable-based dishes, INGREDIENT = Salmon, GENERIC-TERM = Other

The method in this case is described below, see Figure 19.

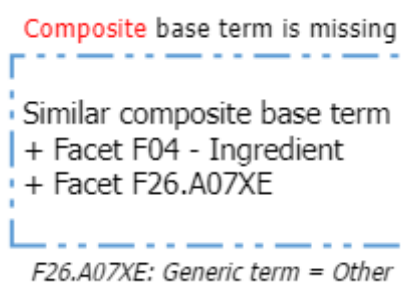


Figure 19: What if the composite *base term* is missing?

3.2 HOW TO CODE MIXED FOOD PRODUCTS?

The concept of convenience food has brought to the market a number of food products ready to use or easy to prepare; among these, many products are actually a mixture of raw commodities or ingredient.

Examples:

Bags of different types of salad ready for consumption or the mixed flours already balanced for specific bakery products (e.g. flour mix for dark bread, flour mix for pizza) or the mixed nuts.

The coding depends on the nature of the components and their balance; the method is described below, see Figure 20.

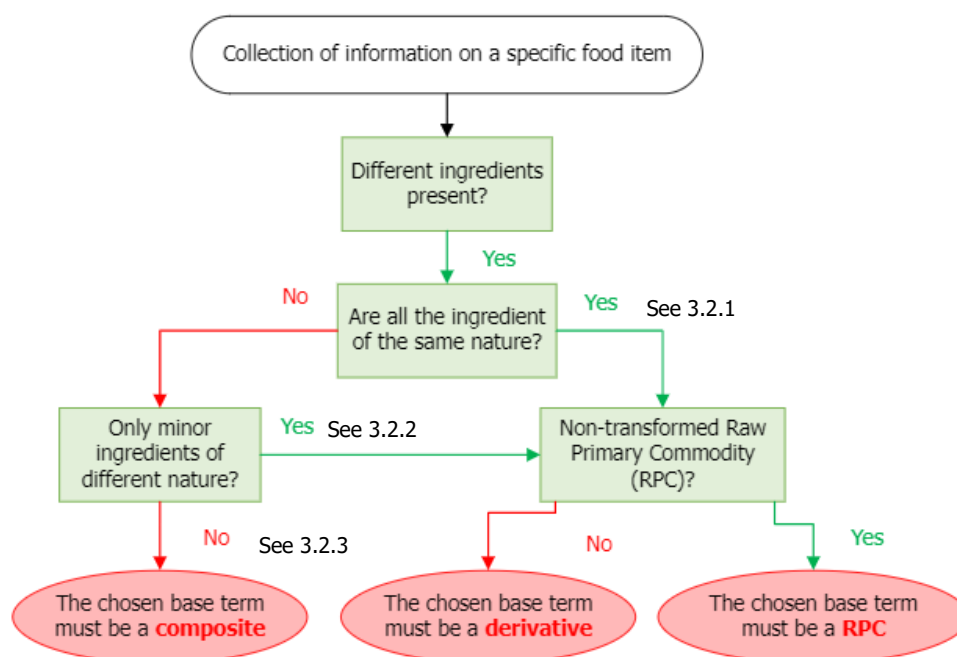


Figure 20: Method to determine the *base term* for mixed food

3.2.1 Different ingredients of the same nature

A **mixture of raw commodities or ingredients of the same nature** should be coded as an RPC or RPC derivative, respectively, using a generic *base term*. The components of the mixture can then be provided by using multiple descriptors of the F27-*Source-commodities* facet (as many as there are components). The components of a mixed food product group are of the same nature when they share the same generic food group.

The starting code would be as follow:

Base term = RPC or RPC derivative
+ Multiple instances of the Facet F27-*Source-commodities*

Example: Mixed leafy vegetables

The components of a salad mixture have the same nature when they are, for example, all belonging to the group “leafy vegetables” [A00KR].

In this case, an appropriate code would use the generic group “leafy vegetables” [A00KR] as the *base term* and different components would be added as multiple instances of the F27-*Source-commodities* facet (not with multiple F01-Source facet descriptors), as follows:

A00KR#F27.A00KV\$F27.A00LB\$F27.A00LG\$F27.A00LN

Leafy vegetables, SOURCE-COMMODITIES = Radicchio, SOURCE-COMMODITIES = Italian corn salads, SOURCE-COMMODITIES = Lollo rosso, SOURCE-COMMODITIES = Roman rocket

3.2.2 Only minor ingredients of different natures

If **almost all components share the same nature**, and if components of different nature are clearly a minority and are in reasonably negligible amounts with respect to the other components, then the components of the mixture which are predominantly of the same nature can be provided using several descriptors of facet F27-*Source-commodities* (as many as there are components), while the minority ingredients of different natures can be provided using facet F04-*Ingredient*.

The starting code would be as follow:

Base term = RPC or RPC derivative
 + Facet F27-*Source-commodities*
 + Facet F04-*Ingredient*

Example: Mixed leafy vegetables with carrots and sunflower seeds

Using the same example as above, but with additional very small amount of carrots and sunflower seeds, coding is done using as *base term* the generic group “leafy vegetables” [A00KR] and adding the different components as multiple descriptors of the F27-*Source-commodities* facet, with the exception of carrots and sunflower seeds, added as minor ingredients with the F04-*Ingredient* facet:

A00KR#F04.A00QH\$F04.A015L\$F27.A00KV\$F27.A00LB\$F27.A00LG\$F27.A00LN

Leafy vegetables, INGREDIENT = Carrots, INGREDIENT = Sunflower seeds, SOURCE-COMMODITIES = Radicchio, SOURCE-COMMODITIES = Italian corn salads, SOURCE-COMMODITIES = Lollo rosso, SOURCE-COMMODITIES = Roman rocket

3.2.3 Different ingredients of different natures

If **different natures are present in the mixture with a substantial balance**, no component is clearly predominant over the others.

Alternatively, **the minority components of the mixture having different nature form the majority of the others are present in non-negligible amount**.

In these two cases, the base term must be of the composite type and the components added with the F04-*Ingredient* facet as follows:

*Base term = Composite
+ F04-Ingredient facet*

Example: Mixed leafy vegetables with carrots and sunflower seeds

Using the same example as above but with more carrots and sunflower seeds, the product is considered as a composite; therefore, the *base term* is a composite term and the components are listed as Ingredient:

A042D#F04.A00QH\$F04.A015L\$F04.A00KV\$F04.A00LB\$F04.A00LG\$F04.A00LN

Mixed vegetable salad, INGREDIENT = Carrots, INGREDIENT = Sunflower seeds, INGREDIENT = Radicchio, INGREDIENT = Italian corn salads, INGREDIENT = Lollo rosso, INGREDIENT = Roman rocket

3.3 HOW TO CODE LIGHT PRODUCTS?

Products traditionally sweetened with sugars (like soft drinks or flavoured yoghurt) may be (or actually are) present on the market as no-sugar or low-sugar alternatives. The same applies to fat content, with reduced fat alternatives of many products available on the market. These products are often referred to as 'light', 'diet' or 'zero'. As a general approach, this concept should be expressed with facets because there are many relevant food groups and creating a 'light' version for each of them would complicate the system. Except for a few specific food groups from previous classifications that are still present: "Diet soft drinks with fruit juice" [A03FV], "Diet soft drinks with flavours" [A03FX], and "Mayonnaise, low fat" [A044Y].

Different descriptors are present in the system in relation to such products, as follows:

- | | |
|---|---|
| <ul style="list-style-type: none"> • F10 - Qualitative-info for fat: <ul style="list-style-type: none"> – A077C low fat (naturally or reduced), – A077D fat free, – A077H skimmed • F07 - Fat-content: <ul style="list-style-type: none"> – A06YH < 0.1 % fat (and similar low fat levels) • F10 - Qualitative-info for energy: <ul style="list-style-type: none"> – A0CRG light • F28 - Process: <ul style="list-style-type: none"> – A07MM lactose reduction | <ul style="list-style-type: none"> • F10 - Qualitative-info for sugars: <ul style="list-style-type: none"> – A077K without added sugar, – A077M low/reduced sugar, – A077L sugar free, – A0B8N low/reduced lactose, – A0CQD lactose free • F08 - Sweetening-agent: <ul style="list-style-type: none"> – Number of descriptors for non-sugar sweeteners (like sugar alcohols and artificial sweeteners). |
|---|---|

In order to avoid a different use of these opportunities by the different coders, a rule for reporting 'light' products is provided:

The standard way of reporting light (or reduced calories) products is therefore by using the following code:

F10.A0CRG "QUALITATIVE-INFO = Light"

All the other descriptors complement this facet by better specifying, depending on the information available, the approach used to reduce calories. **These additional descriptors should not be reported without also using F10.A0CRG "QUALITATIVE-INFO = Light".**

Among the additional descriptors, specifying the sweetener used is particularly important. The descriptor F08.A46M "SWEETENING-AGENT = Artificial sweeteners (e.g., aspartame, saccharine)" should be used if the sweetener is not clarified.

Please note that some words in the descriptors (e.g. reduced) refer to levels defined in specific regulations.

3.4 HOW TO CODE MECHANICALLY TREATED PRODUCTS WITH ALTERED PHYSICAL STATES?

In some cases, the physical state of a food product changes as a result of mechanical action, for example, from solid to paste or from a gel to a liquid. Two aspects have to be taken into account in this context:

- A specific food group exist:
 - The specific food group should be chosen as the *base term* and additional information would not normally be required because the different physical status is implicit.

Example: "A02NQ yoghurt drinks, sweetened and/or flavoured"

The choice of the *base term* is sufficient because the fact that the product, by contrast with 'normal' yoghurt, is liquid is obvious.

- The change can be expressed with facet descriptors:
 - By describing the physical state or by clarifying the altering treatment, via facets F03-Physical-state and F28-Process respectively.

Example:

F03.A0CEE - Spoonable creamy (physical state)	F28.A07KY - Mincing/chopping/cutting (process)
F03.A06JG - Puree-type (physical state)	F28.A0C6N - Pulping/mashing (process)
F28.A0CRK - Stirring (process)	

In general, when there is any doubt, it is always better to give priority to the treatment (F28-Process facet). In few cases, the coder can report both the treatment and the new physical state, if necessary.

3.5 OTHER PARTICULAR RULES OF CODIFICATION

3.5.1 How to code special diet food?

In FoodEx2, there are two ways of coding food item made for special diets:

- There are groups specific for special diets in the classification:
 - o "Food products for young population" [A03PV]
 - o "Food for particular diets" [A03RR] including:
 - "Food for weight reduction" [A03RS],
 - "Dietary foods for special medical purposes" [A03SD],
 - "Food supplements and similar preparations" [A03SJ]
- If there are no terms in the classification, the information can be added as facet with the F23-Target-consumer facet.

3.5.2 How to code vegan or vegetarian foods?

There is the category "Meat and dairy imitates" [A03TD], which includes terms for foods that substitute for meat and dairy products.

In the case of a food that is a vegetarian or vegan alternative, it is recommended to use the F23-*Target-consumer* facet to indicate the target population (vegetarian or vegan). When relevant, the F04-*Ingredient* facet can be used to indicate the plant-based ingredients used.

Example: Vegan cake

A vegan cake is a product in which animal-based Ingredient (milk, butter, and potentially eggs) have been replaced by plant-based substitutes. The "Vegan" descriptor should be used to indicate that the food excludes all animal-derived ingredients.

A possible code is: **A00AN#F23.A1B1Q**
(Cakes, TARGET-CONSUMER = Vegan)

If there is more detail on the ingredient used to replace the animal products, they can be added with facet F04-Ingredient.

3.5.3 How to code a food with range values?

The fat or alcohol content can be specified using the F07-Fat-content (ex: 0.6 % fat) and F11-Alcohol-content (ex: 45 % alcohol v/v) facets, respectively. However, these facets have unique cardinality (see part 2.3.2), meaning only one descriptor from each can be added to the code.

Whether the food label includes a range value, this information should be simplified by choosing the average value as a descriptor. For values above 10%, the available descriptors are integers; therefore, if the decimal is 0.5 or higher, round up to the nearest integer. Otherwise, round down to the nearest integer.

Example: Light margarine with 38–41% fat, enriched with omega-3

The code A039H#F09.A0Evv\$F07.A074D\$F07.A074G for Functional vegetable margarines/fats, FORTIFICATION-AGENT = Omega-3 fatty acids, FAT-CONTENT = 38% fat, FAT-CONTENT = 41% fat, is invalid because it cannot contain two descriptors from the F07 facet.

In this case, the average of the range should be taken: 39.5%. However, since descriptors for values above 10% do not include decimals, the value must be rounded to the nearest integer. Here, because the decimal is 0.5 or greater, we round up.

The valid code is: A039H#F09.A0Evv\$F07.A074F

Functional vegetable margarines/fats, FORTIFICATION-AGENT = Omega-3 fatty acids, FAT-CONTENT = 40% fat.

3.5.4 How to code a food item with an “or” in the name?

In cases where the food item is not specific, such as 'fresh **or** pasteurised semi-skimmed milk', EFSA recommends selecting a general term and adding all the information using facets, even if they are contradictory.

Examples:

Fresh or pasteurised semi-skimmed milk

The general term is: “Cow milk, semi-skimmed (half fat)” [A02LZ], to which the facets can be added F28.A07HV “PROCESS = Pasteurisation”, and F10.A166Y “QUALITATIVE-INFO = Fresh”.

The code is therefore: **A02LZ#F10.A166Y\$F28.A07HV\$**

Cow milk, semi-skimmed (half fat), PROCESS = Pasteurisation, QUALITATIVE-INFO = Fresh

Goose or duck confit

In the same way, the base term has to be general (poultry) and the origin (duck or goose) will be added in the facet string:

A01SN#F06.A06XH\$F27.A01SR\$F27.A01SS\$F28.A07GM

Poultry fresh meat (muscle meat), SOURCE-COMMODITIES = Duck fresh meat, SOURCE-COMMODITIES = Goose fresh meat, PROCESS = Stewing, SURROUNDING-MEDIUM = In animal fat."

3.5.5 How to code sauces?

To choose a *base term* for a sauce, the rule applied must be the main characterizing ingredient as sauces are composite foods, see 2.2.4. In cases where the main characterizing ingredient is not obvious, EFSA created the following prioritisation rule:

Table 15: Top-down priority order of use of food groups for sauces, including alcohol

Food group	Assumed intention while choosing the food	Priority
Meat based...	Source of protein	HIGH 
Fish based...	Alternative source of protein	
Cheese or dairy based...	Source of protein and fat	
Egg based...	Source of protein and fat	
Legume based...	Source of protein and carbohydrates / energy	
Potato based...	Source of carbohydrates / energy	
Cereal based...	Source of carbohydrates / energy	
Fruit based...	Source of micronutrients	LOW
Vegetable based...	Source of micronutrients	
Alcohol based...	Source of energy	

The predefined order facilitates the choice of the base term in case of multiple characterising ingredients without a clear predominance of one of them.

This rule prevails over the content of the Scope Note. If the scope note is in conflict with this rule, then it should not be considered and be reported to EFSA so that they can be corrected.

Example:

Tzatziki sauce: The term Tzatziki [A0FDF] is in the classification. The code should be: **A0FDF**

Pepper sauce:

The term "Pepper sauce" does not exist. The main characterizing ingredient is the pepper.

The code is: **A0ERG#F04.A00HZ**

Herbs/spices sauces, INGREDIENT = Peppers and similar

Bolognese sauce:

The term "Bolognese sauce" does not exist. It is a sauce that contains tomatoes and meat. Two terms can be considered: "Meat sauce" and "tomato-containing cooked sauce". According to the Table 15, meat group are preferred over vegetable group.

The code is: **A16BR#F04.A00HQ**

Meat sauce, INGREDIENT = Tomatoes and similar-

Samurai Sauce:

Similarly, the term "Samurai sauce" does not exist and since it is a mixture of mayonnaise and ketchup, the term used should be "Mayonnaise Sauce" [A00JB], according to the Table 15.

The code is: **A00JB#F04.A044P\$F04.A044X**

Mayonnaise sauce, INGREDIENT = Chili peppers, INGREDIENT = Tomato ketchup and related sauces.

A sauce can be added as a facet with the facet *F04-Ingredient* and a descriptor from the sauce family. If necessary, another descriptor can clarify the content of the sauce if the sauce descriptor is not specific enough.

Example:

Pasta, pesto sauce:

The term Pesto [A044V] exist,

The code is: **A0CDQ#F04.A044V**

Pasta, plain (not stuffed), cooked, INGREDIENT = Pesto

Pasta, pepper sauce:

The descriptor "Pepper sauce" does not exist in the facet *F04-Ingredient*. The sauce descriptor added is Herbs/spices sauces [A0ERG] which is not very specific so the descriptor Peppers and similar- [A00HZ] can be added as ingredient.

The code is: **A0CDQ#F04.A00HZ\$F04.A0ERG**

Pasta, plain (not stuffed), cooked, INGREDIENT = Peppers and similar-, INGREDIENT = Herbs/spices sauces

3.6 EXCEPTIONS: SPICES AND HERBAL INFUSION MATERIALS

Some agriculture products are mostly consumed in a dry or semi-dry form, like spices. To avoid having to create too many instances of a product, there are no terms for the fresh instance of these products.

Spices are therefore considered raw commodities irrespective of the degree of drying.

The actual status has to be reported in case it is not the standard one. Since most of the spices have been subject to a process of drying, in case they are fresh the facet F28.A0C0S - Unprocessed is reported. The same approach also applies for infusion materials such as rooibos.

Until the amendment of 2016, tea was also included in the exceptions but it has been moved back to the standard approach, considering the practices of the tea industry and its market.

4 HOW TO USE OF EFSA TOOLS

The FoodEx2 classification is always evolving. Every year, EFSA releases a new version (major release) of the classification, and several times a year, EFSA publishes minor modifications (minor release) to this version.

The latest version of FoodEx2 while writing this guide was the 15.0, published on January 29, 2024.

Modifications of the classification may affect several important aspects:

- Hierarchies may be added or become obsolete;
- Facets may be added or removed;
- Terms may be added to the classification or become obsolete;
- Hierarchies may be modified:
 - Terms may appear or disappear from a hierarchy (but not from the classification);
 - Relationships between terms may be modified (change of family);
- Properties of terms may change:
 - The type of term may change;
 - The level of detail of the term may change;
 - The implicit facets of the term may change, descriptors may be added or removed.

4.1 THE MTX CATALOGUE AND DATA UPDATES

An excel version of the MTX catalogue (FoodEx2) is available online at this address: <https://zenodo.org/records/10599860>

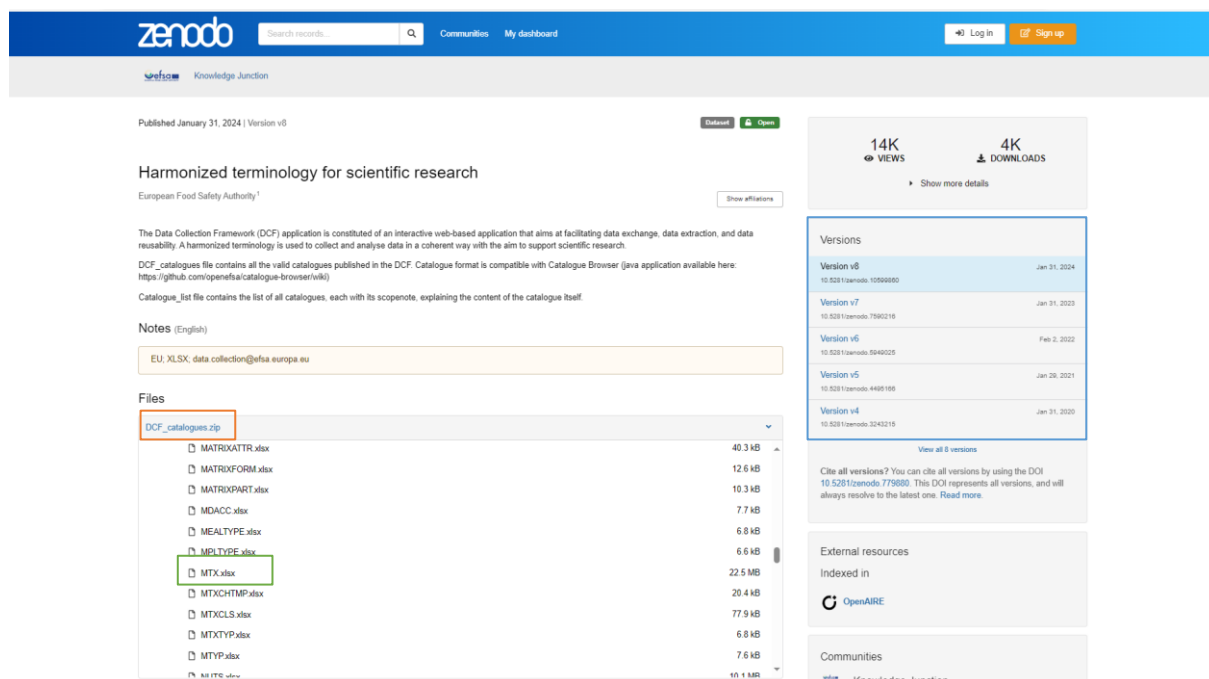


Figure 21: Web page where the Excel version of the MTX catalogue (FoodEx2) is located

On this page, users can choose the **version of the catalogue** (on the right banner) they need and download the **file** containing all the catalogues of the Data Collection Framework of the EFSA. Among these catalogues is the FoodEx2 catalogue, which is called the **MTX catalogue**.

The MTX catalogue can be directly exported from the browser by selecting the “Export Excel” option from the “Tools” menu, as seen in Figure 22.

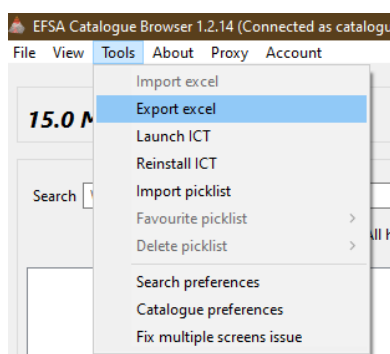


Figure 22: « Tools » menu in the catalogue browser

This Excel file can be useful for developing tools on the classification or for quickly accessing precise data on the terms of the classification. It contains all the terms that have ever been included in the classification, even if they are now obsolete.

In particular, this Excel file can be used to **update databases** in a chosen version of FoodEx2.

The MTX Excel file consists of a table divided in five sheets:

- The first sheet, labelled “catalogue”, contains all the information about the catalogue. Of particular importance are cell J2, which indicates the version of the catalogue, and cell M2, which provides the date from which the catalogue is valid (see Figure 23).

	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	name	label	scopeNote	termCode	termCode	termMinCi	acceptNor	generateM	version	catalogue	lastUpdate	validFrom	validTo	status	depre
1	MTX	MTX (FoodEx2\$[not @\$\$\$@			5	A000A	true	false	14.4	SSD2		2023/10/16		PUBLISHED	0
2															
3															
4															
5															
6															
7															
8															
9															
10															

Figure 23: First sheet of the MTX catalogue

To update a database coded in FoodEx2:

Assuming the version of FoodEx2 that was used is known, the "ValidFrom" column provides the date when the catalogue was released.

- The second sheet, labelled “hierarchy”, contains all the information concerning the hierarchies of the catalogue. Column A include hierarchy codes, meaning the name by which they are referred to as. Column E indicates whether the hierarchy is related to the *base term* (“base”) or the *facets* (“attribute”). Column H provides the date of the last update made in the hierarchy (see Figure 24).

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	code	name	label	scopeNote	hierarchy	hierarchy	version	lastUpdate	validFrom	validTo	status	deprecate	hierarchyGroups	
2	MTX	MTX	MTX (FoodEx2\$[not @\$\$\$@	both	0	15.0		2024/01/	2015/03/		PUBLISHED	0		
3	report	Reporting	Reporting	Reporting	base	1	15.0	2024/01/	2015/03/		APPROVED	0		
4	pest	PesticideR	Pesticide r	Pesticide r	base	2	8.8	2017/06/	2015/03/		APPROVED	0		
5	biomo	Zoonoses	Zoonoses	Zoonoses	base	3	15.0	2024/01/	2015/03/		APPROVED	0		
6	feed	FeedHiera	Feed hiera	Feed hiera	base	4	14.0	2023/01/	2015/03/		APPROVED	0		
7	expo	ExposureE	Exposure l	Exposure l	base	5	15.0	2024/01/	2015/03/		APPROVED	0		
8	vetdrug	VetDrugRe	VetDrugRe	VetDrugRe	base	6	14.4	2023/10/	2015/03/		APPROVED	0		
9	botanic	Botanicals	Botanicals	Botanicals	base	7	15.0	2024/01/	2015/03/		APPROVED	0		
10	part	PartNatur	Part-natur	Part-natur	attribute	8	15.0	2024/01/	2015/03/		APPROVED	0	SSD2\$MTXFACETS	
11	source	Source	Source	Source	attribute	9	15.0	2024/01/	2015/03/		APPROVED	0	SSD2\$MTXFACETS	
12	racsource	SourceCo	Source-co	Source-co	attribute	10	15.0	2024/01/	2015/03/		APPROVED	0	SSD2\$MTXFACETS	
13	process	Process	Process	Process	attribute	11	14.0	2023/01/	2015/03/		APPROVED	0	SSD2\$MTXFACETS	
14	ingred	Ingredient	Ingredient	Ingredient	attribute	12	15.0	2024/01/	2015/03/		APPROVED	0	SSD2\$MTXFACETS	
15	medium	Surroundi	Surroundi	Surroundi	attribute	13	8.8	2017/06/	2015/03/		APPROVED	0	SSD2\$MTXFACETS	
	<	catalogue	hierarchy	attribute	term	releaseNotes		⊕		<				

Figure 24: Second sheet of the MTX catalogue

- The third sheet, labelled “attribute” contains all the information about the facets. Column I specifies the type of the attribute, indicating whether it is a facet (“catalogue”) or something else (“xs:string”). Column D provides details on the scope of each attribute. Column N indicates the cardinality of the attributes. Column S informs about the last update made to the attribute (see Figure 25).

J	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	code	name	label	scopeNote	attributeF	attributeG	attributeH	attributeI	attributeType	attributeJ	attributeK	attributeL	attributeM	attributeN	attributeO	attributeP	attributeQ	attributeR	attributeS	attributeT	attributeU	attributeV	attributeW
2	F02	part	Part-nature	This facet describes the nature of the part	0	true	true	1	catalogue					MTX.part	single	R	false	false	8.10	2017/08/20	2015/03/16		APPROVED
3	F01	source	Source	This facet describes the source of the product	0	true	true	2	catalogue					MTX.source	single	R	false	false	8.10	2017/08/20	2015/03/16		APPROVED
4	F27	resource	Source-commodity	This facet describes the resource of the product	0	true	true	3	catalogue					MTX.resource	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
5	F28	process	Process	This facet allows recording the process of the product	0	true	true	4	catalogue					MTX.process	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
6	F04	ingred	Ingredient	This facet collects ingredients of the product	0	true	true	5	catalogue					MTX.ingred	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
7	F06	medium	Surrounding-medium	This facet is intended for the medium of the product	0	true	true	6	catalogue					MTX.medium	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
8	F08	sweet	Sweetening-agent	This facet allows providing the sweetening agent of the product	0	true	true	7	catalogue					MTX.sweet	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
9	F09	fort	Fortification-agent	This facet allows providing the fortification agent of the product	0	true	true	8	catalogue					MTX.fort	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
10	F10	qual	Qualitative-info	This facet provides some qualitative information about the product	0	true	true	9	catalogue					MTX.qual	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
11	F17	cookext	Extent-of-cooking	This facet describes the extent of cooking of the product	0	true	true	10	catalogue					MTX.cookext	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
12	F26	gen	Genetic-term	This facet allows recording the genetic term of the product	0	true	true	11	catalogue					MTX.gen	single	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
13	F21	prod	Production-method	This facet production method of the product	0	true	true	12	catalogue					MTX.prod	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
14	F18	packformat	Packaging-format	This facet is used for packaging format of the product	0	true	true	13	catalogue					MTX.packformat	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
15	F19	packmat	Packaging-material	This facet is used for packaging material of the product	0	true	true	14	catalogue					MTX.packmat	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
16	F03	state	Physical-state	This facet describes the physical state of the product	0	true	true	15	catalogue					MTX.state	single	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
17	F07	fat	Fat-content	This is a facet with numerical value for fat content of the product	0	true	true	16	catalogue					MTX.fat	single	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
18	F11	alcohol	Alcohol-content	This is a facet containing numerical value for alcohol content of the product	0	true	true	17	catalogue					MTX.alcohol	single	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
19	F12	dough	Dough-Mass	This facet is proposed to record the dough mass of the product	0	true	true	18	catalogue					MTX.dough	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
20	F20	partcon	Part-consumed-amount	When reporting food analysis, this facet allows recording the part consumed of the product	0	true	true	19	catalogue					MTX.partcon	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
21	F22	place	Preparation-recording	This facet allows recording the place of preparation of the product	0	true	true	20	catalogue					MTX.place	single	D	false	false	8.8	2017/06/20	2015/03/16		APPROVED
22	F23	targetcon	Target-consumer	This facet allows recording the target consumer of the product	0	true	true	21	catalogue					MTX.targetcon	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
23	F24	use	Intended-use	This facet allows recording the intended use of the product	0	true	true	22	catalogue					MTX.use	single	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
24	F25	riskingred	Risky-Ingredient	This facet (of specific interest) allows recording the risky ingredient of the product	0	true	true	23	catalogue					MTX.riskingred	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
25	F29	ipurpose	Purpose-of-raising	This facet allows recording the purpose of raising of the product	0	true	true	24	catalogue					MTX.ipurpose	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
26	F30	replev	Reproductive-level	This facet allows recording the reproductive level of the product	0	true	true	25	catalogue					MTX.replev	single	D	false	false	8.8	2017/06/20	2015/03/16		APPROVED
27	F31	animage	Animal-age-class	This facet allows recording the animal age class of the product	0	true	true	26	catalogue					MTX.animage	repeatable	D	false	false	8.10	2017/08/20	2015/03/16		APPROVED
28	F32	gender	Gender	This facet allows recording the gender of the product	0	true	true	27	catalogue					MTX.gender	single	D	false	false	8.8	2017/06/20	2015/03/16		APPROVED
29	F33	legis	Legislative-classes	This facet allows recording the legislative classes of the product	0	true	true	28	catalogue					MTX.legis	repeatable	D	false	false	8.8	2017/06/20	2015/03/16		APPROVED
30	A01	scientificName	Scientific names	Scientific names of the product	0	true	true	29	xs:string	P255													APPROVED

Figure 25: Third sheet of the MTX catalogue

- Most importantly, the fourth sheet, labelled “term”, contains the complete catalogue: It contains 30 columns providing general information on the term along with an additional five columns per hierarchy detailing the position of the term within each hierarchy.
 - Column A provides the unique code of each term (termCode)
 - Columns B, C and D specify the term and its scope (see Figure 26)

	A	B	C	D
1	termCode	termExtendedName	termShortName	termScopeNote
2	A000A	Teff grain		Cereal grains from the plant
3	A000B	Finger millet grain		Cereal grains from the plant
5	A000D	Foxtail millet grain		Cereal grains from the plant
6	A000E	Little millet grain		Cereal grains from the plant

Figure 26: First four columns of the fourth sheet of the MTX catalogue

- Column F indicates when the term was last updated and Column I and J indicate whether the term is still valid (“APPROVED”; 0) or out of use (“DEPRECATED”; 1). See Figure 27.

F	G	H	I	J
lastUpdate	validFrom	validTo	status	depr
2017/07/20	2014/10/30		APPROVED	0
2017/07/20	2014/10/30		APPROVED	0
2019/01/21	2014/10/30	2019/01/21	DEPRECATED	1
2017/07/20	2014/10/30		APPROVED	0
2017/07/20	2014/10/30		APPROVED	0

Figure 27: Columns F to J of the fourth tab of the MTX catalogue

To update a database coded in FoodEx2:

Knowing the date of the FoodEx2 catalogue from which a database was initially coded, one can compare it with the "**LastUpdate**" and "**ValidFrom**" columns to identify which terms have been modified and which ones have been added.

Using the "**deprecated**" and "**status**" columns, obsolete terms can be identified.

- Column M "AllFacets" describes all the implicit facets of the term on top of the *base term*. Some of these implicit facets are inherited from the parent term in the Master hierarchy.

To update a database coded in FoodEx2:

The "**AllFacets**" column might have been modified. In cases where implicit facets have been removed, the user may want to add them back into the code.

- Column U "ImplicitFacets" includes only the implicit facets of the term without the ones inherited from the parent terms.
- Column V "detailLevel" indicates the level of detail of the term (see Table 16).

Table 16: Correspondance entre detailLevel et le niveau de détail d'un terme

detailLevel	Name	Symbol and colour in the FoodEx browser
H	Hierarchy terms	
C	Core terms	
E	Extended terms	
M	Generic terms	
P	Non-specific terms	
F	Facets	

To update a database coded in FoodEx2:

The "**detailLevel**" column may be changed between two versions of FoodEx2. If a *base term* transitions from level C, E, or P to H, M, or F, then it is no longer recommended for coding. Another *base term* should be found.

Conversely, if a term transitions from level H, M, or F to C, E, or P, then this term can be used for coding from now.

- Column W "termType" indicates the type of the term:
 - **n** = Natural sources
 - **r** = Raw Primary Commodities (RPC)
 - **d** = RPC Derivatives
 - **s** = Composite food simple
 - **c** = Composite food aggregated
 - **g** = Broad or mixed
 - **f** = Facets

**The distinction between Simple or aggregated composite food does not imply any specific rules in FoodEx2 coding; it is a distinction present only in the Excel file of the MTX catalogue.*

To update a database coded in FoodEx2:

The **termType** column can be modified between two versions of FoodEx2.

If a *base term* changes type to become a raw commodity, then attention should be paid to the facets indicating the source of the product as well as the process facet, which may require finding another *base term*.

If a *base term* becomes a derivative or a composite food, then attention should be paid to the facets indicating the origin of the product.

- The position of the term in each hierarchy is reported via five columns per hierarchy (xx representing the hierarchy code), including:
 - “xxFlag”: Indicates whether the term is in the hierarchy (1 if yes, None if no);
 - “xxParentCode”: Provides the unique code of the term's parent in the xx hierarchy.

To update a database coded in FoodEx2:

The **xxParentCode** column can be used to identify a term family, its parent and siblings. In the case of a new term, this column enables to look for new child or new sibling of a term, thus providing new ways to encode a food.

- “xxOrder”: Specifies the position of the term in the hierarchy
- “xxReportable”: Indicates if the term is reportable in the hierarchy (1 if yes, 0 if not, None if not in the hierarchy)

To update a database coded in FoodEx2:

The **xxReportable** column should be used to determine if the term is valid within a hierarchy (whether it's a *base term* or a facet).

Caution: In FoodEx2, some terms are "usable" within a hierarchy (xxReportable = 1) even though they are obsolete in the classification (deprecated = 1). In this case, the term cannot be used. This means that it is essential to cross-check the information from both the “**xxReportable**” and “**deprecated**” columns to determine if a term is usable in a hierarchy or not.

- “xxHierarchyCode”: Provides the hierarchy code of the term for the current hierarchy. It is presented as follows, see the example below:

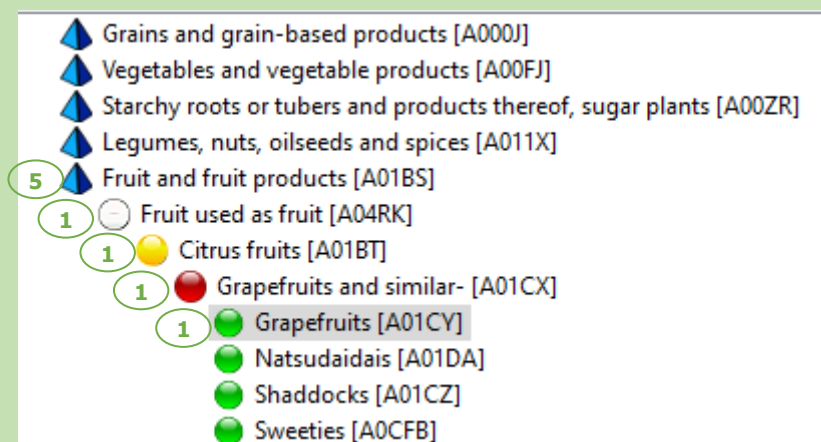
Example: Grapefruit [A01CY]

expoHierarchyCode = Z0005.0001.0001.0001.0001

The code consists of 5 series of 4 digits, indicating that, in the "Exposure" hierarchy, grapefruit is located at the 5th hierarchical level.

To find it in the hierarchy, the coder needs to select:

- The 5th element of the 1st hierarchical level (Fruit and fruit products)
- The 1st of the 2nd level (Fruit used as Fruit)
- The 1st of the 3rd level (Citrus Fruit)
- The 1st of the 4th level (Grapefruits and similar),
- Finally, the term is placed first in the 5th hierarchical level.



Please note that this hierarchy code must not be used as the unique identifier of a base term instead of the termCode in column A. This is because when the MTX catalogue is updated, certain hierarchies may be modified (addition of new groups, movement of a base term to another family, etc.), which will have the effect of modifying this hierarchy code for some base terms.

Nevertheless, this hierarchy code can be very useful for finding all the base terms belonging to a specific family (in a given hierarchy).

Example:

If we are looking for all the foods belonging to the citrus family ("Citrus fruits" [A01BT]) in the Exposure hierarchy, all we need to do is search for all the base terms with a hierarchical code starting with Z0005.0001.0001.

- The fifth and last sheet is a recap of all the modifications made to the catalogue since its last update.

4.2 CREATION OF CODE IN THE CATALOGUE BROWSER

The catalogue browser can help generate codes as well as point out inconsistencies:

- In the catalogue browser, a window (“Describe”) will open with a double click on the selected *base term* as shown in Figure 28. There, the implicit facets are already selected on the left panel:

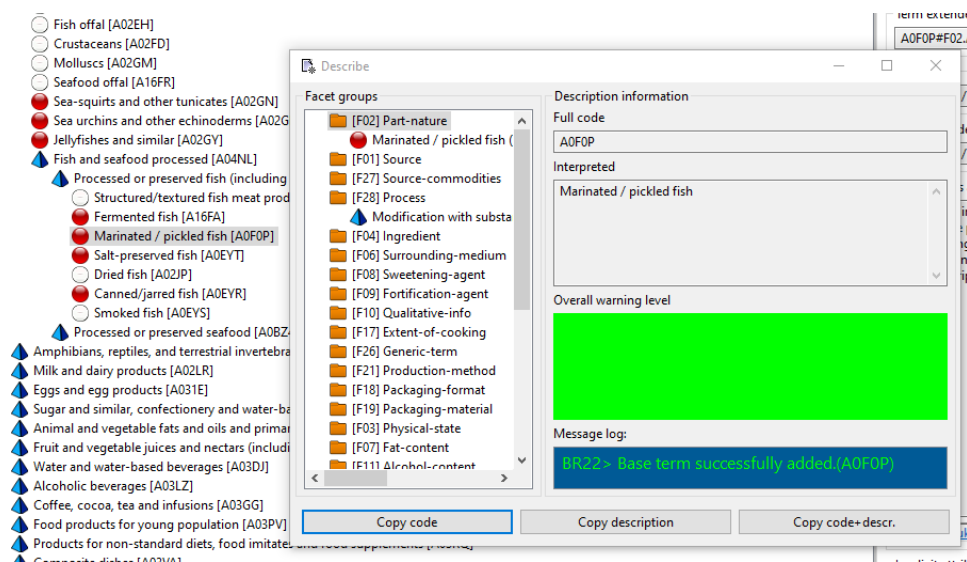


Figure 28: Window “Describe” to code a food in the catalogue browser from a selected base term

- The window “Describe” is split into six parts:
 - The Facet groups: Allows the user to add facets to the FoodEx2 code.
 - The Full code: Displays the FoodEx2 code.
 - Interpreted: Provides the literal translation of each part of the code.
 - Overall warning level: Displays a color-coded indication of the code's validity.
 - Message log: Provides information about the code's validity.
 - Interface: Allows the user to copy the code, the description, or both.

- To add a facet to the code, the user simply needs to double-click on the desired facet. Another window opens as shown Figure 29.

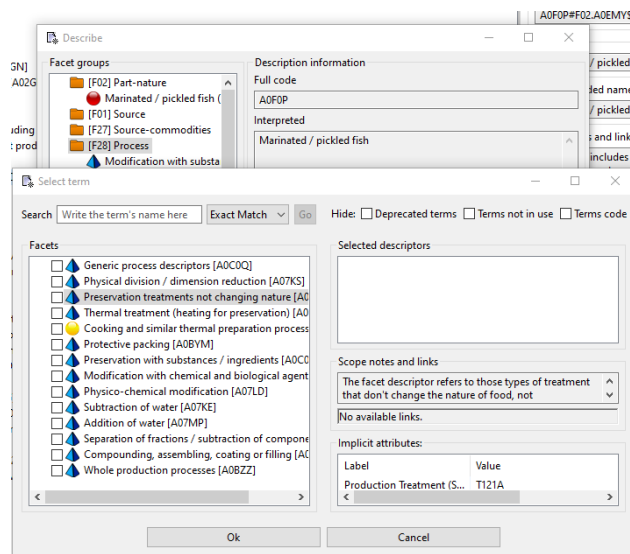


Figure 29: Descriptor choice window after double-clicking on [F28] Process in the Describe window

- When a facet is added, a message appears in the Message Log as shown in Figure 30. Depending on the situation, this message log may or may not indicate a warning. If a warning is mentioned, this does not necessarily mean that the code is wrong, but it should be checked carefully.

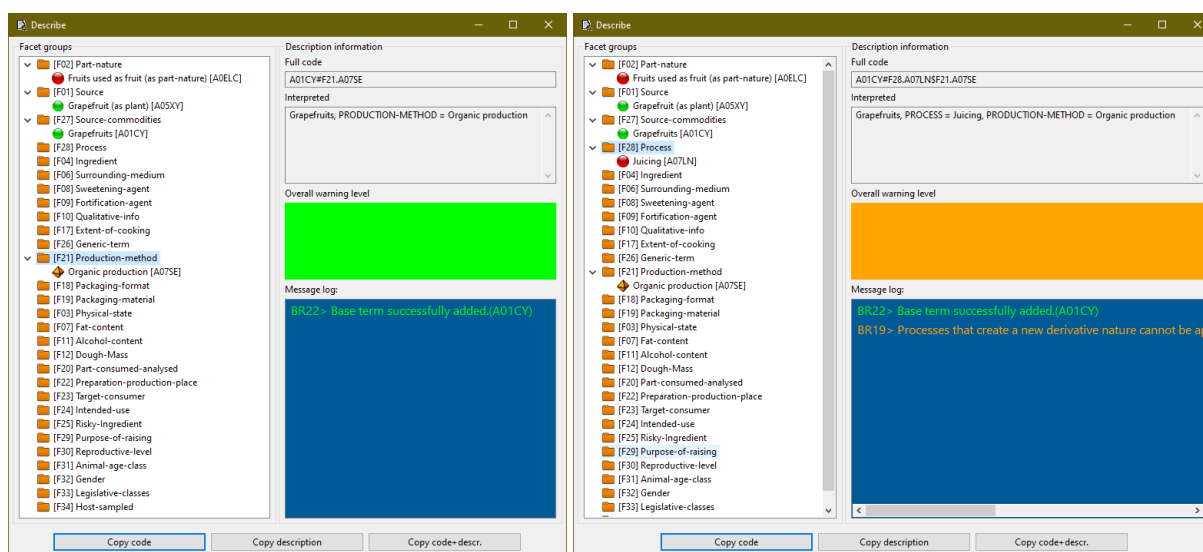


Figure 30: Valid code and not valid code with the warning displayed in the message log

The catalogue browser serves two functions:

- It displays terms in a coherent and usable manner,
- And it operates as a code generator with an integrated verification tool.

However, the code creation function is not very efficient for coding large databases as it only allows working on one code at a time.

4.3 INTERPRETING CHECKING TOOL

An Excel tool was developed and made available in the catalogue browser. It facilitates the verification of the consistency of the proposed FoodEx2 codes with the rules set forth by this classification system, as outlined in this note. To install and run the checking tool, users can select "Install ICT" and "Launch ICT" in the "Tools" menu of the catalogue browser (as seen Figure 31). All the steps required to install and launch the checking tool are available online (<https://github.com/openefsa/Interpreting-and-Checking-Tool/wiki>). A user guide for this tool is also available: <https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/sp.efsa.2019.EN-1727>.

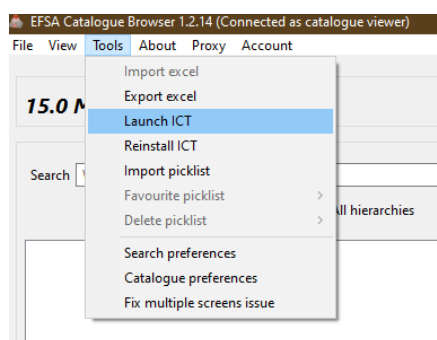


Figure 31: How to launch the ICT via the "Tools" menu

The tool helps performing the quality control steps (see Figure 32). It provides the interpretation of the code creating a field with text explaining the meaning of the code. The tool also records the detailLevel ("Corex") and termType ("Statef") attributes of the *base term* in two added service fields, and recognises if the *base term* is present in the reporting hierarchy and in the exposure hierarchy (reporting Norep and/or Noexp in case of absence).

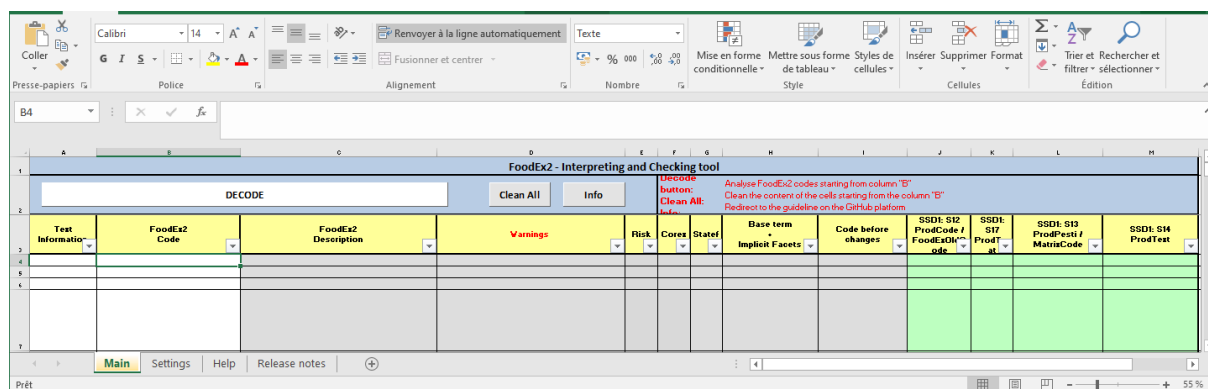


Figure 32: The Checking Tool Interface

All the necessary steps to check the validity of the codes are outlined in ANNEX 2: CHECKING CODES, page 89.

4.4 SMART CODING APP (SCA)

The SCA, developed by EFSA for automatic code generation, isn't recommended for FoodEx2 coding due to several reasons:

- The model is not trained on the latest versions of FoodEx2. The responses it provides may not be up-to-date and, in some cases, incorrect.
- In 2023, EFSA announced that it would no longer maintain the tool.
- This tool does not provide a single FoodEx2 code but rather multiple possibilities.
- To use it and critically evaluate the suggestions provided by the tool, one must have a good understanding of the FoodEx2 nomenclature beforehand.

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ANNEX 1: PRACTICAL EXAMPLES OF FOODEx2 CODES

Despite all the outlined rules explained in this guidance, certain food items remain challenging to code in FoodEx2 because they are specific French dishes or because of translation. Table 17 provides examples of cases that needed discussions to determine an appropriate FoodEx2 code.

The proposed codes are primarily meant to standardise the selection of base terms and some facets. **They are not intended to be exhaustive.** Additionally, these codes are suggested for foods, not for facets. If the food item is an ingredient, the recommended rule is to add the suggested base term with the F04-*Ingredient* facet.

Table 17: Complex codifications cases

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Airelle	<i>Cranberry Cowberry Lingonberry Mountain cranberry</i>	A01FH	Cranberries	FoodEx2 terms for species (both animal and plant) are often very specific, and the English and French terms do not always correspond directly. The French term "airelles" can refer to cranberries as well as cowberries, lingonberries, or mountain cranberries. However, we have chosen to use the term "Cranberries."
Alcool cuit en ingrédient	<i>Alcohol cooked as Ingredients</i>	F10.A0C0V	Alcohol-Free	If there is alcohol in a cooked dish as an ingredient, then this facet needs to be added to tell that the alcohol evaporated during the cooking process.
Anchoïade	<i>Anchovy Paste</i>	A03XV	Fish pâté	There are several categories that include recipes made with fish meat. Anchoïade is similar to items in the "Fish pâté" category due to its texture, consistency, and typical use as a spread.
Baba au rhum	<i>Rum baba</i>	A00AQ# F04.A03PJ	Sponge cake, INGREDIENT = Rum	There is no specific term for "baba au rhum," but the cake part of a "baba au rhum," known as "savarin," is similar to a sponge cake.
Bagel (au pavot) / pain Bagel	<i>Bagel bread</i>	A0BB2	Sandwich bread (hamburger roll-type)	We propose using a common term for all types of sandwich bread to categorise them in the same place.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Barre chocolatée	<i>Chocolate bar</i>	A0EQR# F28.A07HP	Sweet bars and other formed sweet masses, PROCESS = Chocolate coating	In the "Confectionery including chocolate" category, there are several terms that could correspond to the term "chocolate bar." However, there is a specific category, "Sweet bars and other formed sweet masses," that specifically includes the various bars available on the market.
Beurre demi-sel	<i>Slightly salted butter</i>	A039C# F10.A06HX \$ F10.A0CJK	Butter, QUALITATIVE- INFO = With added salt, QUALITATIVE- INFO = Average strength	The facet F10-Qualitative-info can be used to specify the salt "strength" to differentiate between the different types of butter.
Beurre salé	<i>Salted butter</i>	A039C# F10.A06HX \$ F10.A0CJK	Butter, QUALITATIVE- INFO = With added salt, QUALITATIVE- INFO = Strong	The facet F10-Qualitative-info can be used to specify the salt "strength" to differentiate between the different types of butter.
Bière ambrée n.s.	<i>Amber beer n.s</i>	A03MB# F28.A07HC	Beer, PROCESS = Toasting / coffee roasting	The FoodEx2 classification categorises beers based on their type of fermentation; however, colour does not provide information on this criterion. Amber beers are made from slightly roasted malt; we have chosen to code this characteristic to differentiate them from other beers. If a specific beer is mentioned as an example in the label, then refer to that beer's type (Lager, Ale, Wheat beer, or Stout) for coding.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Bière blanche n.s	<i>White beer n.s</i>	A03MM	Wheat beer	The FoodEx2 classification categorizes beers based on their type of fermentation; however, colour does not provide information on this criterion. The vast majority of white beers on the market are Wheat beers; we have chosen to code this characteristic to differentiate them from other beers. If a specific beer is mentioned as an example in the description, then refer to the type of that beer for coding (Lager, Ale, Wheat beer, or Stout).
Bière blonde n.s	<i>Lager</i>	A03MC	Lager beer	The FoodEx2 classification categorizes beers based on their type of fermentation; however, colour does not provide information on this criterion. The vast majority of blonde beers on the market are Lagers; we have chosen to code this characteristic to differentiate them from other beers. If a specific beer is mentioned as an example in the description, refer to that beer's type for coding (Lager, Ale, Wheat beer, or Stout).
Bière brune n.s	<i>Stout, porter</i>	A03MB# F10.A06HX \$ F28.A07HC	Beer, PROCESS = Toasting / coffee roasting, QUALITATIVE- INFO = Strong	The FoodEx2 classification categorizes beers based on their type of fermentation; however, colour does not indicate this criterion. Brown beers are made from heavily roasted malt; we have chosen to code this characteristic to differentiate them from other beers. If a specific beer is mentioned as an example in the description, refer to the type of that beer for coding (Lager, Ale, Wheat beer, or Stout).
Biscotte briochée	<i>Brioche Crispbread</i>	A006M# F12.A008M	Rusk, DOUGH-MASS = Fine yeast sweet dough (brioche)	The facet F12-Dough-mass can be used to indicate the brioche-like quality of the toast.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Biscuit apéritif	Cracker	A06HL# F26.A07XD	Snacks other than chips and similar, GENERIC-TERM = Unspecified	In the case of a generic, unspecified product, we recommend using facet F26- Generic-term to differentiate unspecific products from 'simple' products, depending on the coding requirements.
Cabécou		A02RZ	Cheese, chevre mould ripened	Cabécou is a cheese with a soft, bloomy rind, meaning it is inoculated with <i>Penicillium</i> to form a thin, soft, fluffy, white rind.
Cappuccino décaféiné	<i>decaffeinated cappuccino</i>	A03KH# F04.A03GP \$ F28.A07MN	Coffee drink, cappuccino, PROCESS = Decaffeinating INGREDIENT = Coffee ground, roasted and decaffeinated	Two terms exist: 'Coffee beverage decaffeinated' [A03KF] and 'Coffee drink, cappuccino' [A03KH], but the 'cappuccino' information cannot be easily retrieved through facets. However, the 'decaffeinated' information can be easily added. In this case, it is recommended to use the base term that allows for the easiest addition of facets.
Caillette		A03XB	Pork meat loaf	Caillette is a type of meat and vegetable patty baked in the oven, which can be served as a starter or a main course.
Cake salé	<i>savoury cake</i>	A0CEN	Savoury pies and tarts	A savoury cake is similar to the foods in the 'Savoury pies and tarts' category. Depending on the main ingredients, other terms can be selected, such as A03VV - Meat-based dishes or A03XX - Vegetable-based dishes.
Charcuterie	<i>Charcuterie</i>	A04ND# F27.A\$\$\$\$A ²	Processed whole meat products, SOURCE-COMMODITIES = descriptor	There is no specific term for charcuterie. The term "Processed whole meat products" is most appropriate as it encompasses various types of charcuterie. Its children terms indicate whether the product is raw or cooked.

² F27.A\$\$\$\$A is not a valid FoodEx2 code. It must be replaced by the facet F27 descriptor the most relevant according to the food item to code

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Charolais	<i>Charolais cheese</i>	A02SN	Soft-ripened cheese with natural rind and other soft- ripened cheeses	Charolais is a soft cheese with a natural rind.
Chili con carne	<i>Chili con carne</i>	A03VR# F04.A044C	Beans and meat meal, INGREDIENT = Tomato- containing cooked sauces	Chili con carne is made mainly from beans. Moreover, chili is often accompanied by a tomato sauce
Chocolat avec "saveurs" sans précision si inclusions ou arômes	<i>Flavoured chocolate without precision on the flavouring agent</i>	A034Q	Filled chocolate	When there is no specific information about the type of ingredients added to the chocolate (such as flavouring, inclusion, or filling), always use this term.
Chocolat avec cristaux de sel	<i>Chocolate with salt crystals</i>	A0EQD# F04.A042P (&sons)	Chocolate and similar (&sons), INGREDIENT = Salt	Always use this term if it involves chocolates with salt crystals to differentiate types of chocolate.
Chocolat fourré ou avec inclusion	<i>Filled chocolate or with inclusions</i>	A034Q	Filled chocolate	Always use this term for chocolates with inclusions or fillings to differentiate between types of chocolate. Specify the type of chocolate (dark, milk, white) with the ingredient facet.
Chocolat liégeois	Chocolate mousse topped with whipped cream	A02PV	Dairy desserts spoonable	In recipes for chocolate mousse (chocolat liégeois), there are neither eggs nor gelling agents. <i>Depending on the chosen recipes, the code may vary.</i>
Chocolat simple ou avec arômes	<i>Plain chocolate or with artificial flavouring</i>	A0EQD (& sons)	Chocolate and similar (&sons)	Always use this term if it is plain or flavoured chocolate without inclusions to differentiate between types of chocolate.
Clafoutis		A00BX	Flan tart	Clafoutis is made from a Flan like batter.
Cocktail sans alcool	Virgin cocktail	A03DZ	Soft drinks	Choose this base-term then add the ingredient with the F04-Ingredient

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Cou d'oie farci	Stuffed goose neck	A03VV# F02.A06AH \$ F04.A022F	Meat based dishes, PART-NATURE = Neck (as part-nature), INGREDIENT = Goose, other slaughtering products	Stuffed goose neck is a meat dish consisting of a goose neck filled with stuffing, which can be served with various side dishes. To code this dish, the term "Goose, other slaughtering products" must be chosen to respect the cascade mechanism of FoodEx2
Courge	<i>Marrow, squash, gourd</i>	A0DLV	Pumpkins and similar-	FoodEx2 terms for species (both animal and plant) are often very specific. FoodEx2 differentiates between cucurbitaceae with edible skin and those with inedible skin. We consider that the skin of squash is not edible.
Couscous avec viande	<i>Meat couscous</i>	A03VV	Meat based dishes	The term "couscous" generally refers to the dish itself, which is why we recommend using a term from the "Composite Dish" category. <i>Depending on the chosen recipes, the code may vary.</i>
Couscous de légume	<i>Vegetable couscous</i>	A040M	Pastas and rice (or other cereal) –based dishes	The term "couscous" generally refers to the dish itself, which is why we recommend using a term from the "Composite Dish" category. <i>Depending on the chosen recipes, the code may vary.</i>
Crème caramel		A02PX# F04.A032J	Custard, INGREDIENT = White sugar	"Crème caramel" contains eggs, which contribute to its texture.
Crème dessert bébé semoule	<i>Baby semolina cream dessert</i>	A03RH	Ready-to-eat dairy-based meal for children	A cream dessert is made from dairy products.
Crème dessert pour bébé	<i>Baby cream dessert</i>	A03RH	Ready-to-eat dairy-based meal for children	A cream dessert is made from dairy products.
Crêpe dentelle	<i>Lace crepe</i>	A009X# F04.A00CL	Biscuits, sweet, plain, INGREDIENT = Pancakes	A lace crepe is a biscuit made from a thin crêpe.
Croquant à la noisette	<i>Hazelnut biscuit</i>	A00AE	Biscuit with inclusions, filling or coating;	A hazelnut biscuit is a dry biscuit containing hazelnut pieces.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Croquette de poisson	<i>Fishcake</i>	A03XT	Fishcake	A fish croquette is a dish made from fish, typically consisting of fish mixed with other ingredients, shaped into small patties, and sometimes breaded and fried. This corresponds to the definition of the term "Fish Cake."
Crudités	<i>Raw vegetables/ crudite</i>	A07XJ	Garden vegetables and primary derivatives thereof	To code a general term like "raw vegetable", we recommend using a broad term that encompasses the various vegetables used in crudités.
Eau gazeuse aromatisée	Flavoured carbonated water	A03DY# F28.A07JN	Flavoured bottled water, PROCESS = Carbonating	Add flavours with the F04- <i>Ingredient</i>
Mélange de farine 50% complète 50 % classique	<i>Flour mix, 50% wheat wholemeal flour 50% wheat flour, white</i>	A003X# F10.A06HR\$ F10.A07XK	Wheat flour, QUALITATIVE-INFO = Integral /not refined, QUALITATIVE-INFO = White/refined	This item involves a mix of foods of the same type, but it is not possible to use the facet F27-Source-commodities to indicate the different components since it is not a mix of raw primary commodities, but a mix of the same RPC processed differently. In this case, it is possible to use descriptors of the facet F10-Qualitative-info, even if the information will be contradictory.
Farine de caroube	<i>Carob flour</i>	A01HY# F28.A07LA	Carobs, PROCESS = Grinding / milling / crushing	If the derived term does not exist then the term to choose must be the RPC with a F28- <i>Process</i> descriptor.
Farine de châtaigne	<i>chestnut flour</i>	A0F0N# F03.A07Y2\$ F27.A014J\$ F28.A07LA	Dried nuts/seeds and related flours and powders, PHYSICAL-STATE = Fine powder, SOURCE-COMMODITIES = Chestnuts, PROCESS = Grinding / milling / crushing	If the specific derived term does not exist, a more general term should be used and specified with facets. To assist with this, the implicit facets of existing derived terms can be used as inspiration. In this case, a derived term for 'nut flour' does not exist, but facets F03-Physical-state and F28-Process correspond to the implicit facets of the derived terms for 'flour' (from legumes).

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Farine de lentille	<i>Lentils flour</i>	A185Q# F27.A013Q	Pulses flour, SOURCE- COMMODITIES = Lentils (dry)	To find the correct derived term, first look for the group containing the unprocessed raw material, then search within that group for the corresponding derived term.
Farine de quinoa	<i>Quinoa flour</i>	A04KT# F27.A000R	Cereal and cereal-like flours not separately listed, SOURCE- COMMODITIES = Quinoa	Some categories in FoodEx2 include an 'Other' category. If a precise term does not exist, it is preferable to choose an 'Other' term rather than a higher-level term in the hierarchy.
Farine de toloman	<i>Canna flour</i>	A16RD# F03.A07Y2\$ F27.A010K\$ F28.A07LA	Dried starchy roots and tuber products, PHYSICAL- STATE = Fine powder, SOURCE- COMMODITIES = Canna, PROCESS = Grinding / milling / crushing	If the specific derived term does not exist, a more general term should be used and specified with facets. To assist with this, the implicit facets of existing derived terms can be used as a reference. In this case, a derived term for 'tuber flour' does not exist, but facets F03-Physical-state and F28-Process correspond to the implicit facets of the derived terms for 'flour' (from wheat or legumes).
Feuille de brick	<i>Brick pastry sheet</i>	A16FH# F04.A002R\$ F04.A004F	Phyllo dough, INGREDIENT = Wheat semolina, INGREDIENT = Maize starch	A brick pastry is similar to phyllo dough but includes additional ingredients such as wheat semolina and maize starch.
Feuilleté	<i>Laminated dough</i>	F12.A009L	Laminated doughs	The facet F12-Dough-mass can be used to indicate the laminated dough.
Flammenkueche		A03ZN	Pizza and pizza-like dishes	Flammekueche consists of a bread dough topped with ingredients, similar to a pizza.
Flan aux œufs	<i>Egg flan</i>	A02PX	Custard	Since it consists solely of a mixture made with eggs, it is similar to a dessert cream.
Flan gélifié	<i>Starchy custard</i>	A02PY	Starchy Pudding	The addition of a gelling agent to the mixture makes this food similar to "starchy puddings."
Flan pâtissier avec pâte	Flan	A00BX	Flan tart	There is a term in the FoodEx2 classification.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Flan pâtissier sans pâte	Custard	A02PX	Custard	Since it consists solely of a mixture made with eggs, it is similar to a dessert cream.
Flan salé	<i>Savoury flan</i>	A0CEN	Savoury pies and tarts	A savoury flan is similar to the foods in the 'Savoury pies and tarts' category. Depending on the main ingredients, other terms can be selected, such as A03VV - Meat-based dishes or A03XX - Vegetable-based dishes.
Foie gras de canard brut	<i>Duck, Foie gras (raw)</i>	A01YB# F01.A0CAZ	Duck liver, SOURCE = Duck foie gras animal (as animal)	Please note that this term refers to the raw primary commodity, not prepared and not cooked. If your food is preserved in a can, you should use the term A024B – Canned-tinned meat.
Foie gras de canard (préparé)	<i>Duck, Foie gras (preparation)</i>	A026M# F04.A01YB\$ F10.A0CRE	Liver based spreadable- textured specialties, INGREDIENT = Duck liver, QUALITATIVE- INFO = High fat	If the foie gras is prepared, then a composite term should be used and the duck liver should be added as an ingredient.
Foie gras d'oie (brut)	<i>Goose, Foie gras (raw)</i>	A026N	Foie gras	Please note that this term refers to the raw primary commodity, not prepared and not cooked. If your food is preserved in a can, you should use the term A024B – Canned-tinned meat.
Foie gras d'oie (préparé)	<i>Goose, Foie gras (preparation)</i>	A026P# F04.A026N	Pate, goose liver, INGREDIENT = Foie gras	If the foie gras is prepared, then a composite term should be used and the foie gras should be added as an ingredient.
Foie gras sans précision	<i>Foie gras, unspecified</i>	A026P# F04.A026N	Pate, goose liver, INGREDIENT = Foie gras	Traditionally, foie gras is made from goose liver, and foie gras usually refers more to the prepared product ready for consumption rather than the raw liver. If your food is preserved in a can, you should use the term A024B – Canned-tinned meat.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Fondue savoyarde	<i>Cheese fondue</i>	A16FF	Bread- (or similar products) based dishes	There is no category for cheese based composite dishes in FoodEx2. Savoyard fondue is considered to be a "Bread- (or similar products) based dish" by the EFSA.
Fougasse		A005K	Bread and rolls with special Ingredient added	Fougasse is a type of bread topped with various savoury ingredients.
Gâteau de foie de volaille	<i>Poultry liver cake</i>	A16ER	Offal based dishes	This composite food is an offal-based dish, so an appropriate composite term should be selected.
Ingrédient – céréales	<i>Ingredients cereal</i>	F04.A001X	Mixture of grains	Several descriptors are available, we suggest this one for consistency amongst databases.
Kéfir de fruit	<i>Fruit Kefir</i>	A16YN	Fermented functional drinks (including fermented infusions)	Kefir in French refers to a drink made from fermentation of milk or fruit while the FoodEx2 term refers to only the drink made from the fermentation of milk.
Kéfir de lait	<i>Milk Kefir</i>	A02NV	Kefir	Kefir in French refers to a drink made from fermentation of milk or fruit while the FoodEx2 term refers to only the drink made from the fermentation of milk.
Margarine		A039D	Traditional margarine	The FoodEx2 term "Blended Margarine" refers to specific products, so there is no need to use a vague term for margarine.
Margarine enrichie	<i>Enriched margarine</i>	A039H	Functional vegetable margarines/ fats	The term "Functional vegetable margarines/fats" in FoodEx2 includes fat emulsions and/or fats with added functional substances. All margarines and spreadable fat mixes including functional substances are included.
Mélange fruit secs	<i>Mixed dried fruits</i>	A01QF	Mixed dried fruits	There is a term in the classification ready for use.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Mélange fruit secs et noix	<i>Mixed dried fruits and dried nuts</i>	A06HL# F04.A014C\$ F04.A01QF	Snacks other than chips and similar, INGREDIENT = Mixed dried fruits, INGREDIENT = Tree nuts	It is a mix of food of different natures. According to the rules previously mentioned in part 3.2.3, the base-term must be a composite term and the ingredients added through the F04-Ingredient facet.
Mélanges noix	<i>Nut mix</i>	A0F0N# F27.A\$§\$A ³	Dried nuts/seeds and related flours and powders, SOURCE- COMMODITIES = descriptor	Sold Nuts are usually already dried. For a nut mix, it is a mix of food of the same nature. The term to use should be a derivative as proposed.
Myrtille	<i>Blueberry, bilberry</i>	A04JJ	Blueberries and similar	The French term « Myrtille » refers to either blueberry or bilberry. A general term was preferred since the FoodEx2 classification is quite exhaustive concerning the plants.
Nappage / Sauce au caramel	<i>Caramel topping/ sauce</i>	A046J# F04.A035A	Fudge sauce, Caramel, soft	The term " Fudge sauce" was chosen because it allows the food item to be categorised within a consistent family based on the consumption use.
Nougat blanc	<i>White nougat</i>	A035D# F04.A031T\$ F10.A0F2R	Nougat, INGREDIENT = Hen egg white, QUALITATIVE- INFO = White	White nougat differs from other types of nougat due to the egg whites in its ingredients, which give it its colour.
Nougat chinois	<i>Chinese nougat</i>	A035D# F04.A015K	Nougat, INGREDIENT = Sesame seeds	Chinese nougat is made from caramel and sesame seeds.
Pain à kebab	<i>Kebab bread</i>	A0BB2	Sandwich bread (hamburger roll-type)	We propose using a common term for all types of sandwich bread to group them in the same place.
Pain au lait	<i>Milk roll</i>	A00BT	Brioche type products	Milk roll has a recipe similar to a brioche and is consumed at a similar time.

³ F27.A\$§\$A is not a valid FoodEx2 code. It must be replaced by the facet F27 descriptor the most relevant according to the food item to code

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Pain au levain	<i>Sourdough bread</i>	A0BY0# F04.A05QQ \$ F12.A008K	Leavened bread and similar, INGREDIENT = Sourdough starter, DOUGH-MASS = Naturally leavened bread dough	Sourdough bread is leavened using a mixture of flour and water that has fermented, known as a sourdough starter.
Pain aux céréales	<i>Cereal bread</i>	A005L	Multigrain bread and rolls	"Cereal" is a broad term that can refer to various cereal mixtures; a general term has to be chosen.
Pain aux raisins (viennoiserie)		A00CD# F04.A01ME\$ F04.A02PX	Croissant from puff pastry, INGREDIENT = Dried vine fruits (raisins etc.), INGREDIENT = Custard	In France, the puff pastry "Pain au Raisins" is made from dough similar to a croissant but includes raisins and pastry cream.
Pain de mie	<i>Sandwich bread</i>	A0BB2	Sandwich bread (hamburger roll-type)	In France, the bread "pain de mie" is a type of bread primarily intended for making sandwiches.
Pain grillé	Toast	A006M	Rusk	The term "rusk" refers to a sweet sliced bread browned in an oven.
Pain grillé/toasté suédois	<i>Toasted swedish bread</i>	A0BB2	Sandwich bread (hamburger roll-type)	We propose using a common term for all types of sandwich bread to group them in the same place.
Pain pour panini	<i>Panini bread</i>	A0BB2	Sandwich bread (hamburger roll-type)	We propose using a common term for all types of sandwich bread to group them in the same place.
Pain viennois	<i>Viennese bread</i>	A00BT	Brioche type products	Viennese bread has a recipe similar to a brioche and is consumed in a similar manner.
Pan bagnat		A03ZH	Sandwich with fish and vegetable topping/filling	Pan bagnat is a sandwich made with fish and vegetables.
Pâtisserie	Pastry	A009T	Fine bakery wares	"Pastry" is a very general term, so the chosen term should remain broad.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Pâtisserie orientale	<i>Eastern pastry</i>	A009T	Fine bakery wares	"Eastern pastry" is a very general term and closely related to "pastry." Therefore, the chosen term should remain broad.
Pois Chiche	<i>Chickpea</i>	A013M# F20.A07RB	Chickpeas (dry), PART- CONSUMED- ANALYSED = W/o pods	Chickpeas are generally purchased and consumed without their pods and in a dried form.
Pomme dauphine	<i>Dauphine potato</i>	A011S# F04.A009H	Potato croquettes, INGREDIENT = Choux pastry dough	Dauphine potato are made from a mixture of choux pastry and fried potatoes.
Pomme duchesse	<i>Duchesse potato</i>	A03VD# F04.A031G\$ F04.A0C6H \$ F28.A07GX	Potato based dishes, PROCESS = Baking, INGREDIENT = Potato puree, INGREDIENT = Hen eggs	Duchesse potato are made from a mixture of potato purée and egg yolks, which is then baked.
Pomme noisette		A0BYX	French fries from dough	"Pommes noisettes" is a French dish made from potato purée shaped into small balls and fried in oil.
Potimarron	<i>Onion squash / red kuri squash</i>	A0DLV	Pumpkins and similar-	The French term « potimarron » refers to Onion squash or red kuri squash, both terms belong to the Pumpkins and similar category.
Pralin	<i>Praline</i>	A0F0L# F28.A07LA	Candied or sugar preserved nuts, PROCESS = Grinding / milling / crushing	Praliné is a preparation made from roasted almonds and/or hazelnuts combined with sugar and reduced to a coarse paste.
Préparation pour flan	<i>Flan mix</i>	A0CSM	Cake pre-mixes (dry)	In FoodEx2, there are terms to designate commercial cake mixes available for purchase.
Pulpe de tomate	<i>Tomato pulp</i>	A00ZD	Tomato puree	Without any additional details, we consider tomato pulp to be a homogeneous tomato puree.
Pulpe de tomate avec morceau	<i>Tomato pulp with bits</i>	A00ZC	Preserved tomato, whole or pieces	The description specifies the presence of tomato chunks.
Quatre-quart	<i>Pound cake</i>	A00AP	Plain cakes	A plain quatre-quarts is similar to a plain cake.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Quenelle de poisson	<i>Fish quenelle</i>	A03XJ# F04.A008T\$ F28.A07GH	Fish and seafood-based dishes, PROCESS = Poaching, INGREDIENT = Unleavened doughs	A fish quenelle contains a significant amount of fish. Following EFSA guidelines (see part 2.2.4), we must choose the "Fish based dishes" category and add facets if necessary to specify the dish.
Quenelle de viande	<i>Meat quenelle</i>	A03VV# F04.A008T\$ F28.A07GH	Meat based dishes, PROCESS = Poaching, INGREDIENT = Unleavened doughs	A meat quenelle contains a significant amount of fish. Following EFSA guidelines (see part 2.2.4), we must choose the "Meat based dishes" category and add facets if necessary to specify the dish.
Quenelle nature	<i>Plain quenelle</i>	A03YJ# F04.A008T\$ F28.A07GH	Egg based dishes, PROCESS = Poaching, INGREDIENT = Unleavened doughs	A plain quenelle contains a significant amount of egg. Following EFSA guidelines (see part 2.2.4), we must choose the "Egg based dishes" category and add facets if necessary to specify the dish.
Ratafia		A03NS	Liqueurs	Ratafia refer to an alcoholic beverage made from the maceration of plant materials (such as fruits, grape must, or herbs) in alcohol, and it is synonymous with liqueur.
Ravioli		A0CDP (& sons)	Pasta, filled, cooked (& sons)	Ravioli are stuffed pasta. Depending on the filling, a more specific term may be chosen.
Riz soufflé enrobé de chocolat (confiserie)	<i>Puffed rice coated with chocolate</i>	A034R# F04.A00DR	Chocolate coated confectionery, INGREDIENT = Rice, popped	For confectionery made with puffed rice and coated in chocolate, select a term from the confectionery category.
Sauce carbonara	<i>Carbonara</i>	A044A# F04.A023A	Cream sauce, INGREDIENT = Cured pork fat	The carbonara is mainly made of cream. If there are bacon strip, there are considered as minor ingredients
Sauce chili	<i>Chili sauce</i>	A044P\$ F04.A00JB	Tomato ketchup and related sauces; Chili peppers	Chili sauce primarily consists of tomatoes. Since there are several specific categories for tomato sauces, one should be selected. Chili sauce is a sweet and savoury sauce, which is why the term "Tomato ketchup and related sauces" is chosen.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Saucisson à l'ail	<i>Garlic sausage</i>	A025T	Miscellaneous cooked sausages	A garlic sausage is a preparation made from minced pork meat, seasoned notably with garlic, and cooked during the production process to improve preservation. There is no specific term for it within the "Sliceable or firm cooked sausages" category. However, there is another applicable category that can be used.
Saucisson brioché	<i>Sausage in brioche</i>	A03VV# F04.A00BT\$ F04.A025T	Meat based dishes, INGREDIENT = Brioche type products, INGREDIENT = Miscellaneous cooked sausages	A sausage in brioche is a dish made from a sausage cooked within a brioche. A composite term should be chosen to categorise it.
Saucisson cuit	<i>cooked sausage</i>	A025N	Sliceable or firm cooked sausages	A cooked sausage is a type of sausage where the meat has been cooked during the preparation process to enhance its preservation.
Saucisson de jambon	<i>Ham sausage</i>	A025N	Sliceable or firm cooked sausages	A ham sausage is a pork meat preparation that is cooked during the production process.
Saucisson de noix de jambon	<i>Smoked ham sausage</i>	A024V	Cured ripened raw sausages	A smoked ham sausage is a type of sausage made from pork that has undergone a preservation process.
Saucisson sec	<i>Dried sausage</i>	A024X	Salami-type sausage	A dry sausage is a type of sausage made from fermented and dried meat, often pork. This corresponds to the term "salami-type sausage."
Seitan	<i>Seitan</i>	A03TE# F04.A16GN	Meat imitates, INGREDIENT = Gluten	Seitan is a dish made from wheat or spelt where the starch and bran have been removed, leaving only the gluten. It is commonly used as a meat substitute.
Soubressade		A025B	Pepperoni/pap rika-type sausage	Soubressade is a type of meat preparation that is cooked, spiced, and has undergone a preservation process.

Food Item (French)	Food item (English)	Proposed code	Code translation	Comment
Tapenade		A01QJ# F04.A00GZ \$ F04.A02DD \$F27.A01BP	Fruit or fruit-vegetable puree, <i>SOURCE-COMMODITIES</i> = Table olives, <i>INGREDIENT</i> = Anchovies, <i>INGREDIENT</i> = Garlic	Tapenade is an olive puree mixed with anchovies and garlic.
Telline		A02GZ	Clams, cockles, arkshells	In the FoodEx2 classification, the ISCAAP code for tellins (a broad term encompassing several species) is L.56. The code "Clams, cockles, arkshells" includes all bivalves belonging to the ISCAAP group L.56.
Tripe de bœuf	<i>Beef guts</i>	A16ER	Offal based dishes	Tripe generally refers to a prepared dish made from tripe rather than just the tripe itself.
Zeste de citron	<i>Lemon peel</i>	A01QE# F01.A05VY	Citrus fruit peel, <i>SOURCE</i> = Lemon (as plant)	A specific term for fruit peel exists. A facet F01-Source can be added to specify which fruit the zest comes from.

ANNEX 2: CHECKING CODES

Below are outlined a series of validation steps employing the interpreter checking tool, ICT. These steps encompass the most common issues and queries encountered. Additionally, it is advised to conduct random code validations to uncover any less typical issues.

1 CHECKING FOR FEED OR FOOD

All the feed-related entries in FoodEx2 include in the name the string '(feed)' and are found in the "Feed" hierarchy. A feed code can only be used for feed entries in the occurrence database.

- **Food coded as feed:** First, put a filter keeping only the records containing the string '(feed)' in the interpreted foodEx2 code. Then, put an additional filter for records not containing the string 'feed' in any descriptive text field. The filtered records, if any remains, shall be carefully checked and eventually corrected because they are possibly food items coded with a feed term.
- **Feed coded as food:** put a filter keeping only the records containing the string 'feed' in any descriptive text field. Then, put an additional filter for records not containing the string '(feed)' in the interpreted foodEx2 code. The filtered records, if any remains, shall be corrected because they are feed items coded with a food term.

At the end of this check, the filters have to be removed.

2 CHECKING DETAIL LEVEL: PRESENCE OF HIERARCHY TERMS?

Hierarchy terms (i.e. food groups with detailLevel = 'H') should never be used as *base term*. This aspect is checked by putting a filter for 'H' in the Corex (equivalent to detailLevel). If hierarchy terms are found, this may be due to two reasons:

- 'H' entries due to missing information on the sample:
 - It is suggested to try to the largest extent possible to make assumptions allowing to choose a more detailed term (based on the knowledge of the national market), but in cases when this is not possible, the use of a hierarchy *base term* may become unavoidable. In such case, it should be used a hierarchy term valid in the Exposure hierarchy; by doing this, the data might be eventually considered in some exposure assessments performed at level of broad categories. If a hierarchy term is not present in the Exposure hierarchy, the presence in the Reporting hierarchy shall be checked in the case of chemical occurrence data and the entry shall be corrected in case of absence. For different data collections the use of hierarchy terms from domain-specific hierarchies may be allowed, depending on the particular reporting rules in each domain, if exposure calculations are not involved.
- 'H' entries chosen by mistake and in this case, they shall be corrected.

At the end of this check, the filters have to be removed.

3 CHECKING TYPE OF TERMS: WRONG RPC USED?

A complete check might involve all the possible 'wrong' combinations. A minimum set of checks is suggested here, but it may be further extended.

First of all, a filter for Statef = r should be activated (equivalent to termType). By doing this, only raw commodity *base terms* will be left in the list. Then, the presence of certain facet codes should be checked; the processes for which such a check is suggested are listed Table 18.

Table 18: List of processes for which it is suggested that checks are performed to establish whether or not they should be applied to raw commodities

Code(a)	Process	Code(a)	Process
A0CRH	Seasoning	A0C0A	Grain milling flour production
A07KD	Curing	A0C0B	Starch production
A0C6F	Ripening	A07KG	Drying (dehydration)
A07JP	Preserving by salt	A07JT	Marinating
A07JQ	Preserving by sugars	A07KC	Pickling
A07JR	Candying	A0CQZ	Fermentation
A0C03	Grain milling	A0BYP	Canning/jarring
A0C09	Grain milling early stages	A07JV	Smoking

(a): FoodEx2 code of the Process facet descriptor.

If any of the processes listed above have been applied to a raw commodity *base term*, the code should be checked because there is a high probability that a derivative *base term* exists and should have been used instead.

At the end of this check, the filters have to be removed.

4 CHECKING THE REPORTING OF FLAVOURS

Flavoured products may be ambiguous for the coder, because they may be flavoured with ingredients (e.g. strawberries) with intrinsic flavour not falling under the scope of Regulation (EC) No 1334/2008 or with flavourings (e.g. strawberry flavour) falling under the scope of Regulation (EC) No 1334/2008, or with both.

A check of the ingredients reported in flavoured products shall be performed using a filter in the ICT tool to identify the presence of the string 'flavo' in the interpreted code column. The resulting restricted list must be checked against the content of the text fields describing the food entry; it should be clarified whether it is a regulated flavouring (e.g. strawberry flavour) or a food consumed as such and used as ingredient to provide a specific taste or smell (e.g. strawberries), or both. Incomplete reporting of the actual ingredients used in flavoured food products is very common.

In the context of the FoodEx2 code, all ingredients contributing to the characterising flavour should be mentioned in the FoodEx2 code. If limited information is available at least the ingredient (e.g. strawberries) should be mentioned.

If incomplete reporting is found during this check the entry should be corrected by adding the food used as ingredient.

In conclusion, 'INGR = xyz flavour' should appear as single flavouring ingredient in only those products (e.g. some soft drinks and some yoghurt types) where only regulated flavourings are used and there are no other food Ingredient (fruit, spices, etc.) contributing to the flavour note.

At the end of this check, the filters have to be removed.

5 CHECKING THE REAL NATURE OF INFUSIONS

The use of filters is recommended to identify the presence of 'infus' in the interpreted code and then in the text fields describing the food entry. These two checks must be performed separately. The reason for these filtering steps is that it was often found that infusion ingredients (the powders or dry ingredients) were being reported as infusions (the final reconstituted beverage) or vice-versa. Making this double check and comparing the code with the original information allows such mistakes to be easily spotted.

At the end of this check, the filters have to be removed.

6 CHECKING GRAINS WITH GENERIC PROCESSING

Check the field of the interpreted code with regard to the presence of the string 'grain' and the string 'PROCESS=pr'.

The effect of applying this filter is to isolate foods reported as grains (raw commodity) where a generic process has been reported. It is important to check for terms referring to grains that have been processed, in order to ensure that the processing was not invasive and did not create a new nature; for example, whether the process applied involved cleaning or milling, which would lead to derivatives (as addressed by a previous check). In the latter case (i.e. milling), derivative *base terms* should be chosen (e.g. flour, bran, groats, etc.). This step is specifically suggested for grains, because this problem is frequent. However, similar checks may be performed for all other raw commodities terms, as explained in the next section.

At the end of this check, the filters have to be removed.

7 CHECKING GENERIC PROCESSING APPLIED TO RAW COMMODITIES

This step is similar to the previous one, but, instead of being focused on grains, it extends to all raw commodities.

Two filters should be applied. The first one is to filter for 'r' in Statef (selecting only the raw commodities reported as *base terms*). The second filter is to identify entries in the interpreted code, as previously described, containing 'PROCESS=pr'.

In addition, in this case, the type of process should be checked to ascertain whether or not it could be better defined and whether or not it could eventually lead to a different starting point (*base term*), i.e. a group for derivatives.

At the end of this check, the filters have to be removed.

8 CHECKING INGREDIENT APPLIED TO RAW COMMODITIES OR DERIVATIVES

As a rule, ingredients can be applied as facets to raw commodities or raw commodity derivatives only in the case of minor Ingredient (in small amounts, not clearly quantified and do not change the nature) see part 2.3.5.3, page 42. Some mistakes have been found where the F04-Ingredient facet was used instead of the F01-Source facet or was used to describe mixed raw commodities or derivatives (instead of using the F27-Source-commodities facet). In this last circumstance, multiple F27-Source-commodities facet descriptors should be used instead.

It is suggested that a double filter for 'r' or 'd' in Statef (selecting only the raw commodities and derivatives reported as *base term*) is used. An additional filter should also be used in the interpreted code to filter terms containing 'INGR='.

In this way, all codes where an ingredient is applied to a raw commodity or a derivative are isolated and can be checked one by one, to identify possible errors. It is then easy, by checking the text fields describing the food entry field, to identify the cases where the minor ingredient approach was used (correctly) and those where ingredients were used to describe mixed products (incorrectly) or to specify a source not implicitly included in a specific *base term* (incorrectly).

At the end of this check, the filters have to be removed.

9 CHECKING SOURCES APPLIED TO RAW COMMODITIES

As a rule, mixed raw commodities of the same generic nature should be reported with multiple F27-Source-commodities facet descriptors, not with multiple F01-Source facet descriptors.

In order to identify such cases, a filter for 'r' is used in Statef and a filter for 'SOURCE=' in the interpreted code. The resulting list contains only raw commodities where the F01-Source facet was used and it is easy to identify the multiple use of the facet. An additional way of restricting the list is to add a further filter for the string 'mixed' in the text fields describing the food entry.

At the end of this check, the filters have to be removed.

ANNEX 3: BASE-TERM SUMMARY → RPC, DERIVATED OR COMPOSITE?

How to choose the base-term according to its type?

- **An RPC term** must be chosen if:
 - The food is not processed, [see 2.2.2, page 26](#)
 - The food is a mix of RPC of the same nature, [see 3.2, page 52](#)
- **A derivative term** must be chosen if:
 - One or more processes of the Table 19 was applied to the food. **If several processes are used**, then the choice of the base term must follow the order outlined in the Table 19. [See 2.2.3, page 27](#)
 - The food is a mix of derivatives of the same nature, [see 3.2, page 52](#)

Table 19: Top-down priority order of use of processed/preserved food groups for the choice of the base term in case of multiple treatments

Food group	Underlying Processing	Priority
Puree/textured/structured XXXXX*	Cooking/blanching/separation of parts and mashing or structuring. Eventually associated to canning or packing	HIGH 
Marinated / pickled / fermented XXXXX	Marinating/pickling/fermenting and eventual canning or packing	
XXXXX in vinegar or brine	Pickling/marinating with acid/salt solution and eventual canning or packing	
XXXXX in alcohol or spirits	Preservation with alcohol and eventual canning or packing	
Salted XXXXX	Preserving by salt	
Candied or sugar-preserved XXXXX	Preserving by sugar	
Dried XXXXX	Drying	
Canned/jarred XXXXX	Packing in glass or can or equivalent closed package and thermal treating as needed for preservation	LOW
Smoked XXXXX	Smoking	

*XXXXX replaces the generic name of sections in the classification where the top-down approach applies (e.g. vegetables, meat or fish)

- **A composite term** must be chosen if:
 - The food is a dish. The choice of the base term depends on the main ingredient. If determining the main ingredient is not straightforward, EFSA has defined a prioritisation list seen in Table 20. [See 2.2.4, page 32](#)
 - The food is a mix of derivatives of different natures, [see 3.2.3, page 55](#)

Table 20: Top-down priority order of use of food groups for composite dishes (and similar cases).

Food group	Assumed intention while choosing the food	Priority
Meat based...	Source of protein	
Fish based...	Alternative source of protein	
Cheese or dairy based...	Source of protein and fat	
Egg based...	Source of protein and fat	
Legume based...	Source of protein and carbohydrates / energy	
Potato based...	Source of carbohydrates / energy	
Cereal based...	Source of carbohydrates / energy	
Fruit based...	Source of micronutrients	
Vegetable based...	Source of micronutrients	

The predefined order facilitates the choice of the base term in case of multiple characterising ingredients without a clear predominance of one of them.

ANNEX 4: FACETS SUMMARY → F01, F27 OR F04 FACETS?

When to add a F01-Source facet, a F27-Source-Commodities facet, or a F04-Ingredient facet?

- A F01-Source facet must be added:
 - If the base-term is an RPC, only if the facet helps to better characterise the food. See [2.3.5.1, page 40](#) and [3.1.1, page 48](#)
 - If the base term is a derivative, and there is no sufficiently specific descriptor for the F27-Source-commodities facet, and only if this addition helps to better characterise the food. **Always add the descriptor F26.A07XE "GENERIC-TERM = Others" in this case.** See [3.1.2, page 49](#)
- A F27-Source-Commodities facet must be added:
 - If the base-term is a derivative, only if the facet helps to better characterise the food. See [2.3.5.2, page 41](#) et [3.1.2, page 49](#)
 - If the food is a mix of RPC/derivatives of the same nature:
 - If all the ingredients are of the same nature, they are described by descriptors of the F27-Source-Commodities facet. See [3.2.1, page 53](#)
 - If a few ingredients are of different natures but are minor in the mixture, then the main ingredients are added with the F27-Source-Commodities facet. See [3.2.2, page 54](#)
- A F04-Ingredient facet must be added:
 - If the base-term is a composite, only if the facet helps to better characterise the food. See [2.3.5.3, page 42](#)
 - If the derived food or raw material contains minor or coating ingredients, or flavours. In this case, these ingredients are added with the F04-Ingredient facet. See [2.3.5.3, page 42](#)
 - If the food is a mixture of ingredients of the same nature with minor ingredients of different natures, then these minor ingredients are added with the F04-Ingredient facet. See [3.2.2, page 54](#)
 - If the food is a mixture of ingredients of different natures, then these ingredients are described using the F04-Ingredient facet. See [3.2.3, page 55](#)